



Innovative applications of O.R.

A three-dimensional spatial resource-constrained project scheduling problem: Model and heuristic

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ABSTRACT

For a class of complex engineering projects executed in limited construction sites, spatial resources with three dimensions usually become a bottleneck that hampers their smooth implementation. Serious time-space conflicts frequently occur when multiple activities are carried out in parallel during some time periods. We propose a new project scheduling problem with three-dimensional spatial resource constraints (3D-sRCPSP), and a four-dimensional time-space is adopted to build the model for our 3D-sRCPSP. Firstly, in order to express the impacts of constrained sites on scheduling activities, two types of spatial constraints are refined and modeled: non-overlapping among active subspaces and not exceeding the total space, and then an integer programming model for the 3D-sRCPSP is formalized. Secondly, a novel heuristic algorithm is developed to solve the 3D-sRCPSP, which is embedded with 36 priority rules (PRs) and 3 instructive space allocation strategies. Besides 25 PRs for the traditional RCPSP, 11 new forms of PRs are extracted based on the features of 3D spatial resources. Thirdly, extensive numerical experiments are implemented to validate our model and heuristics. The instances are obtained by configuring the specific parameters of 3D spatial resources for benchmarks from PSPLIB library. One distinctive finding is that some existing PRs that perform well in the traditional RCPSP do not act best in the 3D-sRCPSP. On the other hand, the optimal solutions of very small-scale instances can be obtained by Gourbi solver, but our customized heuristic algorithm is more effective than Gourbi for general instances with medium/large sizes. Overall, the designed heuristics can effectively eliminate time-space conflicts in the planning stage of a project. Finally, as extension studies, the decision tree model is constructed to adaptively select the best PRs for each instance according to the indicators of project instances. These results can help project managers schedule activities and allocate spatial resources more accurately when encountering narrow construction sites.

1. Introduction

The resource-constrained project scheduling problem (RCPSP) has very wide applications in manufacturing, production planning and software development (van der Beek, Souravlias, van Essen, Pruyn, and Aardal, 2024; Hu, Zhang, Wang, Kao, and Ito, 2019). It holds a central position in project management and has given rise to a rich field of research. The traditional RCPSP aims to schedule activities subject to precedence and resource constraints such that the makespan is minimized (Kolisch, 1996a). An idealized assumption in the traditional RCPSP is that construction sites are large enough. However, a class of complex engineering projects with restricted construction sites universally exists in the modern building industry (Dashti, Reza-Zadeh, Khanzadi, and Taghaddos, 2021). For example, metro stations and high-rise buildings in downtown areas, under which the cramped

construction space has become a bottleneck that seriously hinders the efficiency improvement for a project's progress. In some time periods, multiple activities always interfere with each other when they are executed in parallel in the restricted construction site. In fact, space interference is the conflicts among several subspaces occupied by corresponding regular renewable resources (regular resources, in short) required to serve their respective activities, so time-space conflicts are incurred. It is very possible that these conflicts cause serious problems, such as a reduced productivity, project delays and even potential safety accidents. A report from Hong Kong Labor Department indicated that 79% of the deaths in engineering projects are related to space conflicts (Tao, Wu, Hu, and Xu, 2020; Huang and Wong, 2015). Here, we can adopt a real-life complex project to explain such scenes for space

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Fig. 1. The construction scenario of Shanghai Center Tower.

interference and time-space conflicts. The Shanghai Center Tower with 632 meters (m) in height is known as a famous “super project”, whose construction process is very complicated and extremely difficult. Its main structure is a mega framework by the core tube with an outrigger truss. During its construction, four ultra-large tower cranes are installed on the four exterior facades of the core tube. From the 13th floor on, a specific scaffold template equipment is adopted, which is the hydraulic-jacking steel platform (HJSP) supported by the core tube with the 23.35 m height of 5 floors. The intersecting operations in 3D space are very common when working at height. Fig. 1 shows the Shanghai Center Tower under construction. Now, we analyze two representative scenarios of time-space conflicts in this construction site.

Scenario 1: time-space conflicts between four tower cranes and HJSP. During the entire process, each tower crane climbed a total of about 27 times (20–30 meters per time), while the HJSP was raised by 4 meters per 5 days (taking about 6 h). Due to high-altitude operations, the four tower cranes and the HJSP compete for limited space around the core tube. Namely, any one of the four tower cranes and the HJSP cannot be located in the same or overlapping height space around the core tube during some same periods. If the heights of the four tower cranes are not lifted timely, the HJSP will seriously collide with the tower cranes. Meanwhile, the four ultra-large tower cranes busily work in the high sky every day, so any two tower cranes cannot interfere or even collide with each other in their operation space simultaneously. If either of the above two collisions unfortunately takes place, the consequences will be unimaginable. Therefore, there are time-space conflicts between tower cranes and HJSP, as well as among the four tower cranes.

Scenario 2: time-space conflicts during the pouring operation of the large baseplate of the main building. The baseplate with 6 m thickness is as large as the area of 1.6 standard football fields, but the continuous pouring must be strictly controlled to finish within 60 h. In order to execute such a difficult and heavy workload process, a large number of personnel and equipment are deployed. There are over 400 mixing trucks, and the time and the speed of transporting concrete are properly held and coordinated orderly. At the same time, 500 workers and 18 concrete pumps perform simultaneously on site to realize 1000 cubic meters of concrete pouring per hour. In this crowded site, a large number of personnel and equipment must not interfere with each other to ensure the continuous and prompt pouring process. In this context, efficiently tackling time-space conflicts is very important when multiple regular resources work together in very crowded sites.

Similarly to Shanghai Center Tower, many complex engineering projects also encounter the same dilemma of limited construction sites besides high-rise buildings, such as modern prefabricated buildings, deep excavations and underground pipeline networks in cities, etc

(Tian, Zhang, Demeulemeester, and Liu, 2024). However, the traditional RCPSP is based on the assumption of unrestricted construction sites, the space occupied by regular resources required for activities has not been explicitly considered. Obviously, in practical circumstances, space also becomes a scarce resource that must be scientifically allocated. In our study, resources are divided into two categories: regular resources and spatial resources with three dimensions (stereoscopic). Personnel and equipment are the most typical regular resources in project management. Space is used to accommodate equipment and store materials in order to carry out activities (work area) (Dixit, Verma, Raj, and Sharma, 2018; Hu et al., 2019). Apparently, regular resources and spatial resources are occupied simultaneously when executing an activity. Spatial resources can be regarded as a generalization of regular resources; they would degenerate into regular resources if the sizes of multiple dimensions and the shapes of spatial resources are not considered. The size of spatial resources (subspace) for an activity mainly depends on its demands for regular resources. Therefore, we need to allocate both regular and spatial resources efficiently and rationally. To avoid collisions among active activities in the 3D-sRCPSP, the location of each activity must be decided when this activity is executed in a restricted construction site, besides its start time in the traditional RCPSP.

Although fruitful achievements have been obtained for the traditional RCPSP (only containing regular resources), there exists limited literature on the spatial resource-constrained project scheduling problem (sRCPSP). In recent years, only a few scholars dedicated to investigating the two dimensional sRCPSP (2D-sRCPSP) in shipbuilding industry (Lee, Lee, Park, Hong, and Lee, 1997; Zheng, Jiang, Chen, and Liu, 2011). As far as we know, research on quantitatively modeling and solving the 3D-sRCPSP is very scarce. Different from the research on 2D situations, 3D spatial resources should consider not only the volume and the location in restricted construction sites, but also the shape on any projected plane of 2D coordinates. For many large and complex engineering projects, some more realistic scenarios for the joint dimensions of time and space can only be accurately depicted and modeled by resorting to 3D spatial resources. How to effectively handle and avoid the time-space conflicts in the planning stage of these projects is very critical.

To bridge the existing research gap, this study proposes a new project scheduling problem with three-dimensional spatial resource constraints (3D-sRCPSP), also develop the customized heuristics for it. The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 formally describes the formation and constructs the mathematical model for the 3D-sRCPSP. Section 4 is devoted to developing a heuristic algorithm to solve the 3D-sRCPSP. Section 5 conducts extensive numerical experiments to validate the proposed model and designed heuristics. Section 6 draws conclusions and discusses future research.

2. Literature review

There are mainly two streams of related literature on the sRCPSP: the 2D-sRCPSP in shipbuilding projects and workspace interference in engineering projects. Our study involves 3D space allocation, and it is also associated with the 3D bin packing problems (3D-BPP). Therefore, we also summarize the literature related to the BPP. In addition, some efficient PRs for the traditional RCPSP are also reviewed in this section.

2.1. 2D-sRCPSP in shipbuilding projects

The sRCPSP is a class of complex combinatorial optimization problems because scheduling progress and space allocation are treated simultaneously. Lee, Lee, and Choi (1996) firstly put forward the 2D-sRCPSP in shipbuilding projects, and the authors designed a heuristic algorithm based on different positioning strategies to allocate 2D space.

The work of Lee et al. (1996) was followed and extended by subsequent scholars. Koh, Logendran, Choi, and Woo (2011) proposed a 2D packing model and developed a genetic algorithm to solve the spatial scheduling problem in shape-changing mega-blocks. Zhang and Chen (2012) presented a two-stage approach to solve schedule and space allocation problems. Kwon and Lee (2015) also designed a two-stage heuristic algorithm to solve the spatial scheduling problem by using dispatching PRs and a diagonal fill space allocation method. Ge and Wang (2018) incorporated a space allocation strategy into the modified genetic ant colony algorithm and used it to solve irregularly curved block spatial scheduling in shipbuilding projects. Dixit et al. (2018) designed a PRs-based simulation approach to address the block spatial scheduling problem of a shipbuilding project. Hu et al. (2019) developed an advanced guided local search algorithm to solve the RCPSP with 2D spatial resource constraints. Kim, Jeong, and Shin (2020) considered that unnecessary movements frequently occur when ship blocks enter or depart from the stockyard, and developed an asynchronous advantage actor-critic reinforcement learning algorithm to reduce rearrangements.

Overall, a shipbuilding project is always divided into many small blocks, these blocks are assembled into a larger ship, and the space in shipbuilding projects is usually occupied by these blocks. Most studies in shipbuilding projects consider these blocks as 2D spatial resources. However, in our 3D-sRCPSP, the subspaces are occupied by regular resources required for activities, and they should be abstracted into 3D. Obviously, 3D spatial resources are more in line with the realistic scenarios of some complex engineering projects. Moreover, there exist interactions between regular and spatial resources in the 3D-sRCPSP. In addition, three types of algorithms in the literature can be distinguished for solving the 2D-sRCPSP: PRs heuristics, meta-heuristics and deep reinforcement learning algorithms. Currently, the heuristics are more widely adopted because they are more efficient and easily understand. The single PR in heuristics is mostly adopted and the composite PR is rarely involved in the 2D-sRCPSP.

2.2. Workspace interference in engineering projects

Workspace interference usually takes place when multiple activities are executed concurrently in a limited construction site. The reason for the interference is that multiple subspaces occupied by those active activities are physically close to each other. The literature related to workspace interference in engineering projects mainly focus on detecting, managing and eliminating interference automatically through simulation and visualization technology. Akinci, Fischen, Levitt, and Carlson (2002) created a taxonomy of space interference by summarizing their characteristics, and established an approach for analyzing space interference before the construction process. Choi, Lee, Park, Cho, and Kim (2014) categorized workspaces based on their function and mobility, and proposed a framework for the workspace planning process with five stages, including four dimensional building information model (4D BIM) generation, workspace requirement identification, occupation representation, problem identification and problem resolution. Zhang, Teizer, Pradhananga, and Eastman (2015) investigated and prototyped a new BIM methodology for activity-level construction site plan, which is used to improve construction safety. Kassem, Dawood, and Chavada (2015) demonstrated a 4D workspace management tool that complies with Industry Foundation Class, which provides a holistic solution to the approach of workspace management. Dashti et al. (2021) proposed a novel integrated approach based on Discrete Event Simulation, BIM and Rapidly-exploring Random Tree to manage interference automatically.

In general, the existing research on workspace interference in engineering projects mainly involves the following steps: Firstly, they defines the workspace and classifies the type of interference. Then, they obtains or generates the workspace requirement as well as the position information for each activity. Finally, they detects and eliminates the

workspace interference on the BIM visualization platform. Therefore, workspace interference in related studies just refers to the ones that arise during the executions of some activities in parallel. The position information is obtained by the global positioning system or generated by algorithms. Moreover, these gained parameters are entered into the BIM visualization platform to detect and eliminate interference. However, the time-space conflicts in our 3D-sRCPSP are studied from a different perspective, where spatial resources mainly refer to the space occupied by the regular resources needed for activities. Therefore, the location information about the space occupied by regular resources for our 3D-sRCPSP is not known in advance and need to be allocated appropriately. In addition, few studies have been conducted from the perspective of mathematical modeling and obtaining optimal solutions. Meanwhile, we design a new heuristic algorithm with different PRs to provide a basis for resolving time-space conflicts in the next step of BIM development.

Based on the research problems and developed algorithms, the typical literature related to the 2D-sRCPSP in shipbuilding projects, workspace interference in engineering projects, and relevant algorithms are summarized in Table 1.

2.3. 3D bin packing problem and temporal bin packing problem

The 3D-sRCPSP is related to the 3D bin packing problem (3D-BPP) because both of them involve decisions for space allocations. The general 3D-BPP needs to pack a set of known items (boxes) into some large containers with several practical constraints. The objective in the 3D-BPP usually involves maximizing space utilization, minimizing the number of containers or minimizing the variable container height. The 3D-BPP is a strongly NP-hard combinatorial optimization problem (Bischoff and Marriot, 1990; Scheithauer, 1992). Therefore, it is very difficult to solve exactly with larger numbers of boxes (Silva et al., 2019). A variety of approaches was proposed in the literature to solve it, which can be distinguished into seven categories: exact algorithms, constructive heuristics, meta-heuristics, tree-search methods, hyper-heuristics, machine learning techniques and approximation algorithms with theoretical guarantees (Ali, Ramos, Carravilla, and Oliveira, 2022).

Martello, Pisinger, and Vigo (2000) firstly explored to use a branch-and-bound (B&B) algorithm to deal with 3D-BPP. Nowadays, the referenced exact approaches most frequently are mathematical models and combinatorial B&B algorithms. The number of exact algorithms for the 3D-BPP is very small because they can only tackle small-sized problems. Therefore, the existing methods developed for the 3D-BPP mainly focus on heuristic algorithms. Constructive heuristic algorithms aim to create a feasible solution based on a set of PRs, which often form the basis for other more complex heuristics. The meta-heuristic approaches search the whole solution space for the 3D-BPP to gain a satisfactory solution for general-sized problems. Leon, Cueva, Tupia, and Paiva Dias (2019) compared the results of a tabu search algorithm for the 3D-BPP with several meta-heuristics, and concluded that the former could provide reasonable solutions. Nishiyama, Lee, and Mashita (2020) combined the genetic algorithm and physics simulation for a single large object placement problem with multiple constraints. In addition, other heuristic methods are also applied to solve the 3D-BPP or its related problems, such as simulation annealing (Sheng et al., 2017), greedy randomized adaptive search procedure (Correcher, Alonso, Parreño, and Alvarez-Valdés, 2017) and variable neighborhood search (Parreño, Alvarez-Valdés, Oliveira, and Tamarit, 2010). Similarly to other complex problems in operations research, hyper-heuristics and machine learning techniques have also been introduced to tackle the 3D-BPP in recent years (Liu, Zhou, Yang and Zhao, 2023; Que, Yang, and Zhang, 2023), which are all very promising methods for the future. Multiple variants are derived from the general 3D-BPP to cater to different practical scenarios. The pallet loading problem is investigated, which is the 3D-BPP with more practical constraints, such as vertical

Table 1
Literature comparisons.

References	sRCPSP in the shipbuilding projects			Workspace interference in engineering projects		Algorithms				
	2D-SR	3D-SR	RR	QFM	ST	PRH		MH	DRL	EA
						SPR	CPR			
Lee et al. (1996)	✓	–	–	–	–	✓	–	–	–	–
Koh et al. (2011)	✓	–	–	–	–	–	–	✓	–	–
Zhang and Chen (2012)	✓	–	–	–	–	✓	–	–	–	–
Kwon and Lee (2015)	✓	–	–	–	–	✓	–	–	–	–
Ge and Wang (2018)	✓	–	–	–	–	–	–	✓	–	–
Dixit et al. (2018)	✓	–	–	–	–	–	✓	–	–	–
Hu et al. (2019)	✓	–	–	–	–	–	–	✓	–	–
Kim et al. (2020)	✓	–	–	–	–	–	–	–	✓	–
Akinci et al. (2002)	–	–	–	–	✓	–	–	–	–	–
Moon, Kim, Kim, and Kang (2014)	–	–	–	–	✓	–	–	–	–	–
Zhang et al. (2015)	–	–	–	–	✓	–	–	–	–	–
Kassem et al. (2015)	–	–	–	–	✓	–	–	–	–	–
Choi et al. (2014)	–	–	–	–	✓	–	–	–	–	–
Dashti et al. (2021)	–	–	–	–	✓	–	–	–	–	–
This study	–	✓	✓	✓	–	✓	✓	–	–	✓

Legend: 2D-SR: 2D Spatial Resources, 3D-SR: 3D Spatial Resources, RR: Regular Resources, QFM: Quantitatively Formulate Model, ST: Simulation Technology, PRH: Heuristics based on PRs, SPR: Single PRs, CPR: Composite PRs, MH: Meta-Heuristics, DRL: Deep Reinforcement Learning, EA: Exact Algorithms.

support and load bearing and weight limits, etc (Gzara, Elhedhli, and Yildiz, 2020). Typically, the online 3D-BPP has recently been proposed based on some real-life applications, which assumes that only the current items to be packed are observable, but without full knowledge about the characteristics of upcoming items, neither their number nor their sizes (Asta, Özcan, and Parkes, 2016; Yang et al., 2023). Online packing arises in many practical scenarios, such as robotic or automatic container loading in warehouse storage.

The research in the previous paragraph only considers geometric sizes for both boxes and containers, and the time dimension is not considered, so they can be regarded as static packing problems. In comparison to the static issue, a natural generalization of the BPP, called the temporal bin packing problem (TBPP), is addressed under cloud computing environments (Martinovic and Strasdat, 2023). In the TBPP, each item occupies the bin capacity only during a given time window. The TBPP originates from the temporal knapsack problem (Caprara, Furini, and Malaguti, 2013), and it adds a temporal axis to the traditional BPP. The TBPP is called the dynamic packing problem in early publications. The latest papers report that two sophisticated exact algorithms are designed to solve the TBPP, namely a B&B algorithm (Dell'Amico, Jamal, and Montemanni, 2023) and a layer-based combinatorial arc-flow model of manageable exponential size (Martinovic, Strasdat, de Carvalho, and Furini, 2023).

There are obvious differences between the 3D-sRCPSP and the 3D-BPP. A box in the BPP is a tangible entity, but an activity in the 3D-sRCPSP is an abstract expression and refers to a set of some closely related working procedures. The differences between them are carefully summarized as follows:

(1) The 3D-sRCPSP is a dynamic optimization problem with time and 3D sizes in geometry, but the 3D-BPP or BPP is static without the temporal dimension. Although the TBPP is taken as dynamic, the time windows of all packed items are known beforehand. However, for the subspace occupied by the regular resources required by an activity in our 3D-sRCPSP, when to be allocated and where to be positioned in the construction space are just two related decision variables. Therefore, the 3D-sRCPSP is more complex and intractable than both the TBPP and 3D-BPP.

(2) There are no regular resources in the 3D-BPP or BPP, but our 3D-sRCPSP contains the availability of regular resources. More importantly, the complicated interactions exist between regular resources and spatial resources. Besides the limited regular resources, the spatial constraint also restricts and affects the allocations of regular resources. The essence of spatial resource constraints is to coordinate those working regular resources to deal with time-space conflicts. That is to say, because of a limited construction site, it is not possible to place much equipment to accelerate the execution of activities in parallel.

2.4. PRs for the traditional RCPSP

Various PRs for the traditional RCPSP are adopted to resolve the conflicts from competing regular resources among active activities, so these PRs (r-PRs, in short) are summarized. Therefore, in order to more clearly express these r-PRs, we first present the symbols and their meanings in the traditional RCPSP, which are presented in Table 2. Then, we summarize the 25 r-PRs that are effective and frequently used in Table 3.

Depending on the information used by a r-PR, four types of PRs can be distinguished (Klein, 1999): network-based rules (NBR), critical path-based rules (CPBR), regular resource-based rules (RBR) and composite rules (CR). The first six r-PRs use data from a project activity network and discard resource data, including MAI, SAD, MIS, MTS, GRPW and GRPW*. The 7th to 17th r-PRs rely on the results of the forward and the backward recursions of the critical path method (CPM), respectively, including the EST, EFT, LST, LFT, TF, ESTD, EFTD, LSTD, LFTD, TFD and TCI. From the 18th to the 22th, these r-PRs are based on regular resource requirements of activities. They are the GRD, RRCI, WRUP, ACROS and ACTRES. The last type consists of composite rules, which calculate a weighted sum of priority values that combine network-, time- and resource-based rules. They can overcome the shortcomings of depending on only a single type of PR information. For the rules 2–24, ties are broken arbitrarily (randomly). Rule 25 determines priority values randomly and is used purely for comparison purposes.

Even though numerous papers have compared various PRs, there is no agreement on how superior a specific PR is to other rules in different RCPSP settings. Therefore, we need to test which PRs perform best for the 3D-sRCPSP.

In summary, complex engineering projects executed in limited construction sites are universal in the real-life construction industry. However, to the best of our knowledge, the literature about the 3D-sRCPSP is very scarce. In this paper, we first propose a new 3D-sRCPSP, and conduct research using quantitative modeling as well as solving approach. Essentially, the 3D-sRCPSP is an integration of project scheduling problems and space allocation problems for the main purpose of eliminating time-space conflicts. Therefore, this new problem is more complex than the relevant research summarized in the above subsections in Section 2.

Table 2

Symbols and descriptions for the traditional RCPSP with single mode.

Symbols	Descriptions
$G = (N, A)$	Directed acyclic graph of a project's activity network in AoN (activity-on-node);
N	Set of nodes representing all activities, $N = \{1, 2, \dots, J\}$ and $ N = J$;
A	Set of arcs denoting all precedence relations of zero-lag finish-start type;
j	ID-index for activities, $j \in N$ and $j = 1, 2, \dots, J$;
t	Index for time period, $t = 1, 2, \dots, \bar{T}$, where $\bar{T}$ is the upper bound for a project's makespan;
d_j	Duration of activity j in periods;
$P(j)/S(j)$	Set of immediate predecessors/successors for activity j ;
$P(j)^*/S(j)^*$	Set of total predecessors/successors (including direct and indirect) for activity j ;
K	Number of types of renewable resources (regular resources);
k	Index for regular resources, $k = 1, 2, \dots, K$;
r_{jk}	Activity j requires the amount of regular resource type k while being processed;
$rrl_{k\tau}$	Remaining amount of regular resource type k at time τ , $\tau = 1, 2, \dots, \bar{T}$;
R_k	Availability of regular resource type k ;
CPM	Critical path method (without the limits of regular resources);
ES_j/EF_j	Earliest start/finish times of activity j based on the classic CPM;
LS_j/LF_j	Latest start/finish times of activity j based on the classic CPM;
TF_j	Total float of activity j based on the classic CPM, $TF_j = LS_j - ES_j = LF_j - EF_j$;
s_j/f_j	Start/finish times of activity j satisfying the availability of all regular resources;
ES_{j-t}^d/LS_{j-t}^d	Updated earliest/latest start times of activity j along resource allocation progress at time instant t , $j \in N \setminus \{i s_i \leq t\}$, namely, j belongs to the set of unscheduled activities at t
EF_{j-t}^d/LF_{j-t}^d	Updated earliest/latest finish times of activity j along resource allocation progress at time instant t , $j \in N \setminus \{i s_i \leq t\}$, namely, j belongs to the set of unscheduled activities at t ;
π_j	Number of paths placed after activity j in a activity network;
$\prod_{jh}$	Set of activities on path h placed after activity j , $h = 1, \dots, \pi_j$.

3. Formulations and models

3.1. Problem description

The 3D-sRCPSP inherits the basic framework from the traditional RCPSP, and extends the latter from the perspective of differences in resource attributes. The symbols and meanings for the traditional RCPSP are defined in Table 2, so this section will not elaborate on these in detail. In the 3D-sRCPSP, it is necessary to consider the space sizes and the time periods occupied by regular resources required for an activity. In order to express the structure of 3D space, the cartesian coordinate system with three axes is adopted, and its visualization is shown in Fig. 2. The three axes correspond to the length direction $X(l)$, the width direction $Y(w)$ and the height direction $Z(h)$ of a limited construction site, respectively. On X, Y and Z axes, we indicate the directions of left–right, front–back and up–below, respectively. Please note that the Y axis is represented by a dashed arrow, which is different from that for X and Z axes. Therefore, the direction of the Y axis towards the coordinate origin (O) is called backward, and the direction away from O is called forward. In addition, the time axis $T(t)$ is an extension of the cartesian coordinate system, and it is also placed within it, thus forming a four-dimensional structure. Fig. 2 indicates that the time axis is depicted as a dashed line, whose gradient color ranging from light to dark signifies the progress of gradually scheduling activities within a project as time elapses. A restricted construction site is taken as a large cuboid (denoted by V) with three dimensions of length (L), width (W) and height (H). The subspace occupied by active activity j is abstracted into a small cuboid (denoted by v_j) with constant length (l_j), width (w_j) and height (h_j). The vertex (x_{jt}, y_{jt}, z_{jt}) in the left-back-below corner of v_j is considered as its reference point (rp). When v_j is active at time instant t ($s_j < t \leq f_j$), it can be accurately positioned in V by its rp . Obviously, the 3D coordinates (x_{jt}, y_{jt}, z_{jt}) of the rp are just one of the two types of decision variables (location and start time of v_j). Similarly, in a 2D plane, the vertex at the left-below of a rectangle is used as its rp to locate its specific location. The proximity degree of an object (2D or 3D) to O is defined as the distance between its rp and O . The symbols and meanings for the 3D-sRCPSP are summarized in Table 4.

In this study, we must consider the sizes of cuboid v_j occupied by active activity j in three geometric dimensions (length, width and height) rather than only involving its volume value. The sizes of this cuboid can be derived from the number of regular resources required for this

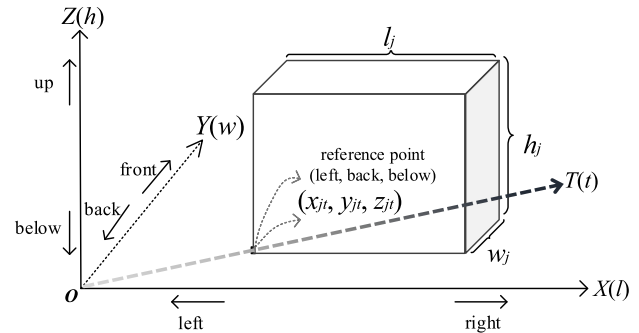


Fig. 2. The visualization of 3D cartesian coordinate system and time axis.

activity. In a cramped construction site, multiple regular resources are working together during some periods, and serious time-space conflicts will likely be incurred. From the view of project activities, the principle of time-space conflicts can be interpreted: the locating of the cuboids occupied by various regular resources required for activities will affect the scheduling of those activities (time dimension). Conversely, the sequence of scheduling activities also impacts the occupation and the release of spatial resources (space dimension). The essence of time-space conflicts lies in a limited construction site, and the space becomes a scarce resource (spatial resource) during a project's execution. Besides 3D spatial resources, we also must consider the precedence relationships between all pairs of activities and the availability of regular resources in a project planning stage. At the same time, the interactive relations between regular resources and spatial resources must be handled. In general, we aim to get optimal or satisfactory schedule schemes that eliminate time-space conflicts, and improve the efficiency as well as the safety of some large and complex engineering project executions in limited construction sites. Obviously, the 3D-sRCPSP involves both time-space schedule and space allocation decisions.

Fig. 3 presents an example of the time-space visualization, which is used to illustrate the scheduling process of time-space. When $t=1$, the cuboids occupied by active activities j_1 and j_2 are v_1 and v_2 , respectively. At $t=8$, j_2 is completed and v_2 is freed, then the active activities convert to j_1, j_3, j_4 and j_5 . Along a project's progress, the space layout status is changed and updated dynamically. Four activities

Table 3
25 PRs for regular resources in the traditional RCPSP (r-PRs).

Types	Num.	Acronyms: Descriptions	Formulas	References
NBR	1	MAI: minimum activity ID	$\text{Min}\{j\}$	Olaguibel and Goerlich (1989)
	2	SAD: shortest activity duration	$\text{Min}\{d_j\}$	Davis and Patterson (1975)
	3	MIS: most immediate successors	$\text{Max}\{ S(j) \}$	Chen, Demeulemeester, Bai, and Guo (2018)
	4	MTS: most total successors	$\text{Max}\{ S(j)^* \}$	Kolisch (1996a)
	5	GRPW: greatest rank positional weight	$\text{Max}\{(d_j + \sum_{i \in S(j)} d_i)\}$	Valls, Perez, and Quintanilla (1992)
	6	GRPW*: old greatest rank positional weight	$\text{Max}\{(d_j + \sum_{i' \in S(j)^*} d_{i'})\}$	Cooper (1976)
CPBR	7	EST: minimum earliest start time	$\text{Min}\{ES_j\}$	Klein (1999)
	8	EFT: minimum earliest finish time	$\text{Min}\{EF_j\}$	Klein (1999)
	9	LST: minimum latest start time	$\text{Min}\{LS_j\}$	Davis and Patterson (1975)
	10	LFT: minimum latest finish time	$\text{Min}\{LF_j\}$	Kolisch (1996a)
	11	TF: minimum total floats	$\text{Min}\{TF_j\}$	Davis and Patterson (1975)
	12	ESTD: minimum dynamic ES time	$\text{Min}\{ES_{j-t}^d\}$	Klein (1999)
	13	EFTD: minimum dynamic EF time	$\text{Min}\{EF_{j-t}^d\}$	Klein (1999)
	14	LSTD: minimum dynamic LS time	$\text{Min}\{LS_{j-t}^d\}$	Klein (1999)
	15	LFTD: minimum dynamic LF time	$\text{Min}\{LF_{j-t}^d\}$	Klein (1999)
	16	TFD: minimum dynamic TF	$\text{Min}\{TF_{j-t}^d\}$	Klein (1999)
	17	TCI: time criticality index	$\text{Max}\{((\text{Max}\{TF_j\} - TF_j)/\text{Max}\{TF_j\})\}$	Dixit et al. (2018)
RBR	18	GRD: greatest resource demand	$\text{Max}\{(d_j \sum_{k=1}^K r_{jk})\}$	Ulusoy and Özdamar (1989)
	19	RRCI: regular resource criticality index	$\text{Max}\{\sum_{k=1}^K (r_{jk}/R_k)\}$	Boctor (1990)
	20	WRUP: weighted resource utilization and precedence	$\text{Max}\{(w S(j) + (1-w)\sum_{k=1}^K (r_{jk}/R_k))\}, w=0.5$	Ulusoy and Özdamar (1989)
	21	ACROS: activity resource controls	$\text{Max}\{(\text{Max}\{\sum_{i \in \prod_{jh}} \sum_{k=1}^K (r_{ik}/R_k) h=1, \dots, \pi_j\})\} =: arc_j$	Elsayed and Nasr (1986)
	22	ACTRES: activity resource	$\text{Max}\{(\text{Max}\{\sum_{i \in \prod_{jh}} \sum_{k=1}^K d_i (r_{ik}/R_k) h=1, \dots, \pi_j\})\} =: act_j$	Elsayed and Nasr (1986)
CR	23	TIMROS: based on the LST and ACROS	$\text{Max}\{(w((LS_j - LS_j)/(LS_j - LS_1)) + (1-w)(arc_j/arc_1))\}, w=0.5$	Bedworth (1973)
	24	TIMRES: based on the LST and ACTRES	$\text{Max}\{(w((LS_j - LS_j)/(LS_j - LS_1)) + (1-w)(act_j/act_1))\}, w=0.5$	Bedworth (1973)
–	25	RAD: randomly determine priority values	$\text{Max}\{\text{rand}()\}$	Klein (2000)

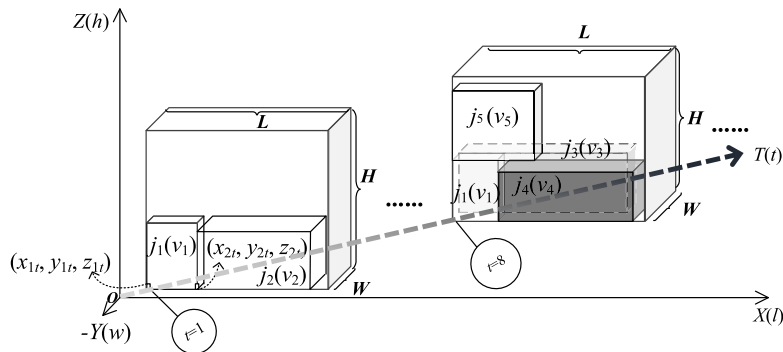


Fig. 3. An example of the time-space visualization.

Table 4

Symbols and descriptions for the 3D-sRCPSP based on the traditional RCPSP (in Table 2).

Symbols	Descriptions
Parameters	
V	Total space of a limited construction site;
L, W and H	Length, width and height values of V , respectively;
v_j	Cuboid occupied by active activity j , and is abstracted as a small cuboid, $\forall j \in N$;
l_j, w_j and h_j	Length, width and height values of v_j occupied by active activity j ;
X, Y and Z	Three coordinate axes in length, width and height directions, respectively;
T	Time axis along a project's progress;
US_{j-xy}	Projection of v_j on the X - Y plane (2D);
US_{j-xz}	Projection of v_j on the X - Z plane (2D);
US_{j-yz}	Projection of v_j on the Y - Z plane (2D);
M	Infinitely large positive number;
0-1 auxiliary variables	
$\alpha_{ijt}, \beta_{ijt}$ and γ_{ijt}	Description of the projection state between v_i and v_j in different planes formed by X, Y and Z axes, respectively;
φ_{jt}	If activity j is being executed at time instant t , then φ_{jt} equals 1, 0 otherwise;
Decision variables	
δ_{jt}	If activity j starts at time instant t , then δ_{jt} equals 1, 0 otherwise;
x_{jt}, y_{jt} and z_{jt}	At time instant t , three coordinate values of rp for activity j in V . Namely, the left-black-below corner of v_j when v_j is active.

are executed in parallel at $t=8$, and we take activities j_1 and j_5 as realistic scenarios to explain time-space conflicts. Supposing that j_5 is painting the exterior walls of a large building, and j_1 involves trimming and widening the road very close to this building. Correspondingly, v_5 is the cuboid for the moving scaffolds required by j_5 , and v_1 is the cuboid for the large paving machines used by j_1 . We must rush the schedule in the limited construction site, so these two activities have to be carried out concurrently. In this situation, the limited space is competed by those working regular resources (moving scaffolds and paving machines). Obviously, in the outer area of the lower floors of this building, conflicts are likely to arise between the two cuboids for moving scaffolds (j_5) and for several large paving machines (j_1). Therefore, when a large number of equipment, machinery and other regular resources are working together during some periods, they must not conflict to ensure a project's smooth progress. It is very crucial and challenging to allocate proper cuboids at the right moment for these working regular resources under the limited total space, so that project managers can avoid time-space conflicts and improve construction efficiency.

3.2. Precedence and regular resource constraints

As mentioned above, 3D-sRCPSP is an extension and innovation of the traditional RCPSP in 3D spatial resources. The precedence and the regular resource constraints are inherited from the traditional RCPSP. Thus, we begin by presenting the model of the traditional RCPSP as follows (Klein, 1999):

$$\text{Minimize } \sum_{t=ES_j}^{LS_j} \delta_{jt} \times t \quad (1)$$

$$\delta_{jt} = \begin{cases} 1, & \text{if activity } j \text{ starts at time } t, \forall j \in N, t = ES_j, ES_j + 1, \dots, LS_j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

$$\sum_{t=ES_j}^{LS_j} \delta_{jt} = 1, \forall j \in N \quad (3)$$

$$\sum_{t=ES_i}^{LS_i} \delta_{it} \times (t + d_i) \leq \sum_{t=ES_j}^{LS_j} \delta_{jt} \times t, \forall i, j \in N, i \in P(j) \quad (4)$$

$$\sum_{j=1}^J r_{jk} \sum_{\tau=\max\{ES_j, t-d_j\}}^{\min\{LS_j, t-1\}} \delta_{j\tau} \leq R_k, \forall j \in N, k = 1, 2, \dots, K; t = 1, 2, \dots, \bar{T} \quad (5)$$

The objective function Eq. (1) expresses the start time of the dummy end activity J , which realizes to minimize a project's makespan. Eq. (2) defines 0-1 decision variables δ_{jt} . Eq. (3) ensures that only one start

moment of activity j is adopted from the interval between ES_j and LS_j . The precedence constraints in Eq. (4) indicate that activity j can be started only if all immediate predecessors are finished. The regular resource constraints in Eq. (5) show that at each time period, the regular resource requirements of activities in progress cannot exceed its availability for each resource type. In addition, ES_j , LS_j , EF_j and LF_j of activity j in the model can be obtained by the forward and the backward recursions from the classic CPM, respectively (Klein, 1999).

3.3. Representation and formulation of 3D spatial resource constraints

How to quantitatively express the impact of a restricted construction site on some activities' schedule is the most important work to construct the model of the 3D-sRCPSP. Two types of constraints can be categorized: one implies that there exists no overlap between any active cuboids, the other indicates that the sum of volumes of all cuboids occupied by those active activities at any period cannot exceed the total space. Later, we will clearly explain the specific meanings of these two types of customized constraints for the 3D-sRCPSP.

We first introduce relevant variables used in the model of the 3D-sRCPSP. A specific 0-1 auxiliary variable φ_{jt} is introduced in Eq. (6) to indicate whether activity j is executed at time instant t . Eq. (7) expresses that activity j is scheduled at time instant t and ensures that the processing of activity j must take d_j consecutive periods since its start. Obviously, Eq. (7) can also express the process of dynamically allocating space.

We use activity j to elaborate on Eq. (7). If activity j has $ES_j=1$, $LS_j=5$ and $LF_j=9$, its actual start time $s_j=5$, $\delta_{j5}=1$, $d_j=4$ and $f_j=9$. Therefore, at $t=5$, $\varphi_{j5} = \delta_{j1}+\delta_{j2}+\delta_{j3}+\delta_{j4}=0$, v_j is not occupied by activity j ; while at $t=6$, $\varphi_{j6} = \delta_{j2}+\delta_{j3}+\delta_{j4}+\delta_{j5}=1$; at $t=7$, $\varphi_{j7} = \delta_{j3}+\delta_{j4}+\delta_{j5}=1$; at $t=8$, $\varphi_{j8} = \delta_{j4}+\delta_{j5}=1$; at $t=9$, $\varphi_{j9} = \delta_{j5}=1$. Thus, when t goes from 6 to 9, v_j is occupied by activity j and all φ_{jt} are equal to 1. When $t > 9$, activity j is finished, v_j will be freed and reallocated to other eligible activities.

$$\varphi_{jt} = \begin{cases} 1, & \text{if activity } j \text{ is executed in } V \text{ at time instant } t, \forall j \in N, t = 1, 2, \dots, \bar{T} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

$$\varphi_{jt} = \sum_{\tau=\max\{ES_j, t-d_j\}}^{\min\{LS_j, t-1\}} \delta_{j\tau}, \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (7)$$

3.3.1. Non-overlapping among active cuboids

When multiple activities are executed in parallel in V and each activity occupies its cuboid, the non-overlapping constraint means that there must be no intersection between any two active cuboids. If we

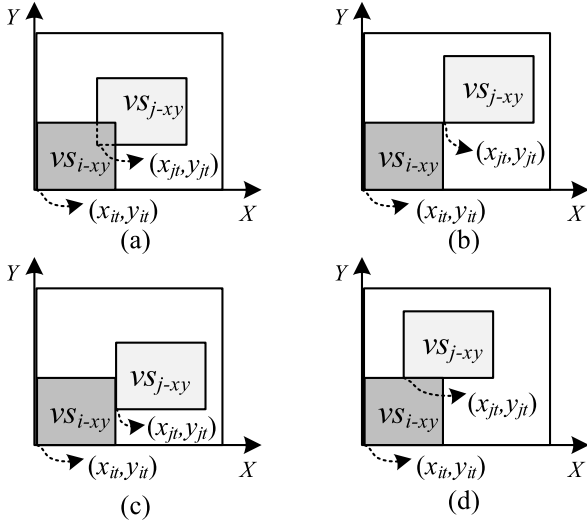


Fig. 4. The possible projection states of v_i and v_j on the X - Y plane.

want to judge whether two cuboids overlap or not in 3D space, the projections of each cuboid on three 2D planes are observed. The three 2D planes are formed by coordinate axes X , Y and Z , which are X - Y , Y - Z and X - Z , respectively. Correspondingly, the projection of cuboid v_i (v_j) on the X - Y plane is denoted as VS_{i-xy} (VS_{j-xy}), on the Y - Z plane as VS_{i-yz} (VS_{j-yz}) and on the X - Z plane as VS_{i-xz} (VS_{j-xz}). The interference states of the two cuboids in 3D space can be determined by the number of overlapping projections (NOP) of the two cuboids on the three 2D planes (X - Y , Y - Z and X - Z).

Supposing that v_i is closer to the coordinate origin than v_j . For any two cuboids v_i and v_j in 3D space, we adopt the X - Y plane to illustrate the possible states of their projections, which is shown in Fig. 4. Apparently, an overlap occurs between VS_{i-xy} and VS_{j-xy} only if both $x_{it} + l_i > x_{jt}$ and $y_{it} + w_i > y_{jt}$ are satisfied. Non-overlapping happens when at least one of $x_{it} + l_i \leq x_{jt}$ and $y_{it} + w_i \leq y_{jt}$ holds. Fig. 4a shows the overlapping situation on the X - Y plane, while Fig. 4b, Fig. 4c and Fig. 4d exhibit three different non-overlapping scenarios. The possible projection states of v_i and v_j on the Y - Z or on the X - Z are the same as those on the X - Y . On three 2D planes, the NOP between two cuboids can be determined, which is used to express their interference states in 3D space. Therefore, we will formally analyze the overlapping and the non-overlapping situations for any two cuboids in 3D space by introducing a proposition.

Proposition 1. Whether two cuboids overlap or not in 3D space mainly depends on the following properties:

(a) If the NOP between v_i and v_j on the three 2D planes (X - Y , Y - Z and X - Z) is less than or equal to 1, then there exists no overlap between them in 3D space.

(b) If v_i and v_j overlap in 3D space, then their projections will overlap on three 2D planes.

We employ the proofs by contradiction to demonstrate Proposition 1a. Firstly, we assume that cuboids v_i and v_j do not overlap in 3D space, but the NOP is greater than 1. Since we suppose that the NOP is greater than 1, there must exist overlapping projections on at least two 2D planes. Let us consider that these overlapping projections occur on X - Y and X - Z planes, obviously, v_i and v_j will overlap in X , Y and Z axes. However, if two cuboids do not overlap in 3D space, they must be completely separated at least one dimension of X , Y and Z axes. Therefore, this contradicts our assumption. Based on the principle of proof by contradiction, our assumption that the NOP is greater than 1 is incorrect. Therefore, we conclude that if two cuboids do not

overlap in 3D space, the NOP is less than or equal to 1. Similarly, Proposition 1b can also be proven.

There are a total of four non-overlapping situations:

(1) If the projections of v_i and v_j exist the overlap only on the X - Y plane ($NOP=1$), it means that $x_{it} + l_i > x_{jt}$, $y_{it} + w_i > y_{jt}$ and $z_{it} + h_i \leq z_{jt}$ should be satisfied;

(2) If the projections of v_i and v_j exist the overlap only on the X - Z plane ($NOP=1$), the inequality $x_{it} + l_i > x_{jt}$, $z_{it} + h_i > z_{jt}$ and $y_{it} + w_i \leq y_{jt}$ should be satisfied;

(3) If the projections of v_i and v_j exist the overlap only on the Y - Z plane ($NOP=1$), then $y_{it} + w_i > y_{jt}$, $z_{it} + h_i > z_{jt}$ and $x_{it} + l_i \leq x_{jt}$ should be satisfied;

(4) If the projections of v_i and v_j do not overlap on three 2D planes ($NOP=0$), then $x_{it} + l_i \leq x_{jt}$, $y_{it} + w_i \leq y_{jt}$ and $z_{it} + h_i \leq z_{jt}$ should hold true.

Therefore, when the NOP is less than or equal to 1, at least one of the inequalities $x_{it} + l_i \leq x_{jt}$, $y_{it} + w_i \leq y_{jt}$ and $z_{it} + h_i \leq z_{jt}$ should be satisfied. Obviously, v_i and v_j do not overlap in 3D space. In order to clarify the projection states (on three 2D planes) between v_i and v_j at time instant t , three auxiliary variables α_{ijt} , β_{ijt} and γ_{ijt} are adopted, whose roles are formally defined and given in Eqs. (8)–(10).

$$\alpha_{ijt} = \begin{cases} 1, & \text{if on } X\text{-}Y \text{ and } X\text{-}Z \text{ planes formed along } X \text{ axis, the} \\ & \text{projection of } v_i \text{ is closer to the coordinate} \\ & \text{origin than that of } v_j \text{ and no overlap exists between their} \\ & \text{projections at time instant } t \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

$$\beta_{ijt} = \begin{cases} 1, & \text{if on } X\text{-}Y \text{ and } Y\text{-}Z \text{ planes formed along } Y \text{ axis, the} \\ & \text{projection of } v_i \text{ is closer to the coordinate} \\ & \text{origin than that of } v_j \text{ and no overlap exists between their} \\ & \text{projections at time instant } t \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

$$\gamma_{ijt} = \begin{cases} 1, & \text{if on } X\text{-}Z \text{ and } Y\text{-}Z \text{ planes formed along } Z \text{ axis, the} \\ & \text{projection of } v_i \text{ is closer to the coordinate} \\ & \text{origin than that of } v_j \text{ and no overlap exists between their} \\ & \text{projections at time instant } t \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

Therefore, we can describe the non-overlapping constraints between any active cuboids (v_i and v_j) at time instant t by designing the mathematical formulas, which are shown in Eqs. (11)–(20).

$$x_{it} - x_{jt} + l_i - M(1 - \alpha_{ijt}) \leq 0 \quad (11)$$

$$y_{it} - y_{jt} + w_i - M(1 - \beta_{ijt}) \leq 0 \quad (12)$$

$$z_{it} - z_{jt} + h_i - M(1 - \gamma_{ijt}) \leq 0 \quad (13)$$

$$\alpha_{ijt} + \alpha_{jit} \leq \varphi_{it} \quad (14)$$

$$\alpha_{ijt} + \alpha_{jit} \leq \varphi_{jt} \quad (15)$$

$$\beta_{ijt} + \beta_{jit} \leq \varphi_{it} \quad (16)$$

$$\beta_{ijt} + \beta_{jit} \leq \varphi_{jt} \quad (17)$$

$$\gamma_{ijt} + \gamma_{jit} \leq \varphi_{it} \quad (18)$$

$$\gamma_{ijt} + \gamma_{jit} \leq \varphi_{jt} \quad (19)$$

$$\alpha_{ijt} + \alpha_{jit} + \beta_{jit} + \beta_{jit} + \gamma_{ijt} + \gamma_{jit} \geq \varphi_{it} + \varphi_{jt} - 1 \quad (20)$$

Eqs. (8) – (20) are all satisfied $\forall i, j \in N$, $i \neq j$, $i \notin P(j)^*$ and $j \notin P(i)^*$, $t = \min\{ES_i, ES_j\}$, $\min\{ES_i, ES_j\} + 1, \dots, \max\{LF_i, LF_j\}$

(21)

Eqs. (11)–(20) express the non-overlapping constraints between any v_i and v_j . In addition, Eqs. (11)–(20) also present other specific cases between v_i and v_j by resorting to a large enough integer M , which

includes that only one of v_i and v_j is active, or both of them are not active. If one of the inequalities is satisfied in Eqs. (11)–(13), it means that the projections of v_i and v_j do not overlap on any two 2D planes. If two or three inequalities are satisfied in Eqs. (11)–(13), it represents that the projections of v_i and v_j do not overlap on three 2D planes. Therefore, when $NOP \leq 1$, we need to ensure that at least one of the inequalities is satisfied in Eqs. (11)–(13) through Eqs. (14)–(20). Eq. (21) gives the ranges of i, j and t in Eqs. (8)–(20).

3.3.2. Not exceeding the total space (V)

On the main premise that there exists no overlap among active cuboids, then for each period t ($t = 1, 2, \dots, \bar{T}$), all active cuboids cannot surpass the boundary of a limited construction site during a project's progress, which is termed as the sum of volumes of all active cuboids cannot exceed V . Namely, the three boundaries (L , W and H) of V cannot be surpassed, which is depicted by the 3D coordinates of the left-back-below corner of any active cuboid and its sizes (l, w and h) in three directions.

The auxiliary variable $\varphi_{jt}=1$ if v_j is occupied by activity j , otherwise $\varphi_{jt}=0$. A large enough integer M ($M > 0$) is introduced to assist in formulating the constraints. Therefore, if $\varphi_{jt}=1$, in the length direction (X), the sum of x_{jt} and l_j is not greater than L of V , which is shown in Eq. (22). Similarly, Eqs. (23) and (24) indicate the same meanings in the width (Y) and in the height (Z) directions of V , respectively. At the same time, $x_{jt} \geq 0, y_{jt} \geq 0$ and $z_{jt} \geq 0$, which are shown in Eqs. (25)–(27). Conversely, if $\varphi_{jt}=0$ ($t \leq s_j$ and $t > f_j$), $x_{jt} + l_j - M \leq L$, $y_{jt} + w_j - M \leq W$ and $z_{jt} + h_j - M \leq H$ are satisfied in Eqs. (22)–(24). In addition, $x_{jt} \geq L$, $y_{jt} \geq W$ and $z_{jt} \geq H$ are also fulfilled in Eqs. (25)–(27). Eqs. (28)–(30) indicate that the location of v_j cannot be changed during its execution.

$$x_{jt} + l_j - M(1 - \varphi_{jt}) \leq L, \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (22)$$

$$y_{jt} + w_j - M(1 - \varphi_{jt}) \leq W, \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (23)$$

$$z_{jt} + h_j - M(1 - \varphi_{jt}) \leq H, \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (24)$$

$$x_{jt} \geq L(1 - \varphi_{jt}), \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (25)$$

$$y_{jt} \geq W(1 - \varphi_{jt}), \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (26)$$

$$z_{jt} \geq H(1 - \varphi_{jt}), \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (27)$$

$$x_{jt} - L(1 - \varphi_{jt}) = x_{j(t+1)} - L(1 - \varphi_{j(t+1)}), \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (28)$$

$$y_{jt} - W(1 - \varphi_{jt}) = y_{j(t+1)} - W(1 - \varphi_{j(t+1)}), \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (29)$$

$$z_{jt} - H(1 - \varphi_{jt}) = z_{j(t+1)} - H(1 - \varphi_{j(t+1)}), \forall j \in N, t = ES_j, ES_j + 1, \dots, LF_j \quad (30)$$

3.4. 3D-sRCPSP model

Based on the two types of customized constraints for 3D spatial resources (non-overlapping among active cuboids and not exceeding the total space), the integer programming model for the 3D-sRCPSP can be formulated as follows.

$$\begin{aligned} & \text{Minimize} \quad \sum_{t=ES_j}^{LS_j} \delta_{jt} \times t \\ & \text{s.t.} \quad (2)–(30) \end{aligned}$$

The objective function of the 3D-sRCPSP is to minimize a project makespan. The 3D-sRCPSP model includes the constraints in Eqs. (2)–(5) from the traditional RCPSP and also the 3D spatial resource constraints in Eqs. (6)–(30). Two kinds of decisions in the 3D-sRCPSP model are involved and they interact with each other: 1) how to allocate space for each activity (spatial component); 2) how to schedule each activity (temporal component).

Based on the careful analysis of mathematical formulas, the characteristics of the 3D-sRCPSP model are analyzed as follows: The 3D-sRCPSP is strongly NP-hard. The NP-hard of the 3D-sRCPSP can be demonstrated by converting the 3D-sRCPSP to the traditional RCPSP after some relaxation. Obviously, if we do not consider the 3D spatial resource constraints in Eqs. (6)–(30), the model of the 3D-sRCPSP can degenerate into the traditional RCPSP. The latter is a well-known strongly NP-hard problem (Blazewicz, Lenstra, and Kan, 1983). Therefore, the 3D-sRCPSP is strongly NP-hard.

The 3D-sRCPSP extends and innovates the traditional RCPSP. Moreover, the 3D-sRCPSP is more complex and intractable than the traditional RCPSP because of interactions between regular resources and spatial resources, as well as time and 3D space. At the same time, an enormous number of binary variables are adopted to model the 3D-sRCPSP. Therefore, commercial optimization solvers can only solve very small-scale instances, we can find the optimal solution for small-scale instances of 15 activities within acceptable time. It is necessary to design an efficient heuristic algorithm to solve the large-scale instances for the 3D-sRCPSP.

4. A specific heuristic algorithm for 3D-sRCPSP

In this section, the general framework of the heuristic algorithm based on some PRs for time-space schedule and space allocation is presented in Section 4.1. PRs and space allocation strategies play a crucial role in our designed algorithm. Therefore, Section 4.2 shows the new forms of PRs based on spatial resource features (s-PRs). In Sections 4.3–4.5, we describe three space allocation strategies, respectively.

4.1. The general framework of heuristics based on PRs for time-space schedule and space allocation

Generally, PR and schedule generation scheme (SGS) are core components in the heuristic algorithm to solve the traditional RCPSP. A PR is used to select the activity that has to be scheduled first. An SGS defines how a feasible schedule can be constructed by continuously assigning the start time for each activity (Kolisch, 1996b; Chen et al., 2018; Liu, Zhou, Lim and Hu, 2023). More specifically, an SGS aims to generate a feasible schedule by extending a partial schedule (i.e. a schedule where only a subset of all activities has been assigned a start time) in a stage-wise fashion (Kolisch, 1996b). In each stage, an SGS forms the set of all schedulable activities. An SGS is distinguished into two types: serial schedule generation scheme (SSGS) and parallel schedule generation scheme (PSGS). The difference between SSGS and PSGS lies in how they advance decision stages during scheduling. SSGS is activity-oriented and PSGS is time-oriented. Obviously, time-oriented PSGS is more suitable for the time-space features of the 3D-sRCPSP. However, in our study, dynamically allocating space and scheduling activities are conducted simultaneously along a project's progress. This process is always accompanied by the coupling relations between time and 3D space, and between regular resources and 3D spatial resources. Therefore, the traditional SGSs are improper for solving our 3D-sRCPSP, it must be modified to deal with the 3D-sRCPSP by introducing space allocation strategies.

Some related studies about space allocations in the 3D-BPP indicated that the heuristic algorithm based on placement points can produce better results (Crainic, Perboli, and Tadei, 2008; Fekete and Schepers, 2004). However, placement points in the 3D-BPP are usually static, so the existing heuristic algorithm based on placement points in the 3D-BPP is unsuitable for the dynamic 3D-sRCPSP. Therefore, a specific heuristic algorithm based on multiple types of PRs and various space allocation strategies is designed for the 3D-sRCPSP, in which scheduling activities and allocating space simultaneously along time axis dynamically. Our heuristic algorithm extends the existing ones based on placement points for the 3D-BPP from three aspects. Firstly, several new forms of PRs based on spatial resource features

Table 5
The definition of several sets.

Sets	Definitions
PP^t	Set of placement points at time instant t , including two scenarios: three placement points ($PP_{jx}^t \leftarrow$ right-back-below vertex of v_j , $PP_{jy}^t \leftarrow$ left-front-below vertex of v_j and $PP_{jz}^t \leftarrow$ left-back-up vertex of v_j) are generated along X, Y and Z axes when v_j is placed in V ; one placement point occupied by v_j is freed when activity j is finished. In addition, $PP^t[p][1]$ is the first element of placement point p . In other words, it is also the coordinate of placement point p in X axis. $PP^t[p][2]$ and $PP^t[p][3]$ are the coordinates in Y and Z axes, respectively.
RPP_t	Set of residual placement points at time instant t .
A_t	Set of ongoing activities at time instant t .
C_t	Set of finished activities at time instant t .
E_t	Set of eligible activities that only satisfy precedence constraints at time instant t .
RE_t	Set of rigorously eligible activities that satisfy both precedence and regular resource constraints at time instant t .
TE_t	Temporary set composed of the activities in RE_t that cannot satisfy spatial constraints at time instant t .
APP	Set of placement points of those activities that have been scheduled.
ST/FT	Set of start/finish times of those activities that have been scheduled.

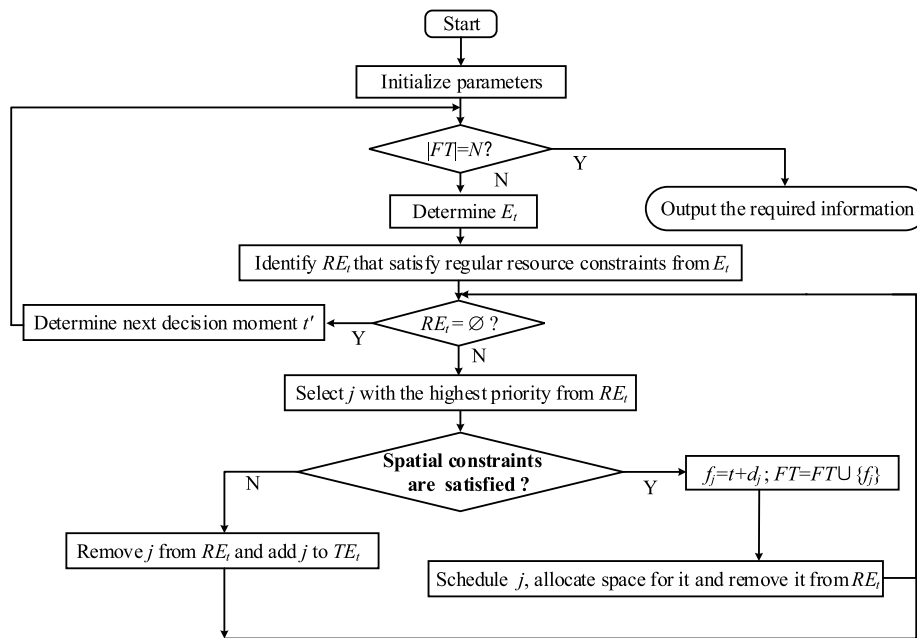


Fig. 5. The general framework of our designed heuristic algorithm.

are proposed. Secondly, the dynamic generation strategy for placement points is designed, in which new placement points are continuously created along a project's progress. Finally, the dynamic search strategy for placement points and the secondary allocation strategy for residual placement points are devised to use space efficiently. Moreover, we can obtain each activity's start time and its location during it is executed.

The definitions of the sets used in our heuristic algorithm are listed in Table 5. The general framework of our designed heuristic algorithm is shown in Fig. 5. The focus in this framework is how to allocate space dynamically at each decision moment. The pseudo-code of the heuristic algorithm is shown in Algorithm 1, which can demonstrate the detailed steps of this algorithm.

Obviously, our designed algorithm (PR-PSGS-SAS) involves the PR (including r-PR and s-PR), the PSGS and the space allocation strategies (SAS), which can achieve the time-space schedule and space allocation efficiently. Specifically, a thorough analysis of the new forms of PRs and the three space allocation policies (the dynamic generation strategy for placement points (PPDGS), the dynamic search strategy for placement points (PPDSS) and the secondary allocation strategy for residual placement points (RPPSAS)) is essential in the following sections.

4.2. New forms of PRs based on spatial resource features

Before describing the newly designed PRs based on spatial resources, we will give the formal definition of the set of rigorously eligible activities.

Definition 1. The set of rigorously eligible activities (RE_t) is composed of activities that satisfy the precedence and the regular resource constraints. RE_t is mainly obtained by selecting activities that satisfy the regular resource constraints from E_t . However, activities in E_t are those that only satisfy the precedence constraints. E_t consists of three components: remaining activities in $E_{t'}$ at the previous decision time t' , unscheduled activities in the temporary set $TE_{t'}$ that cannot satisfy the spatial constraints at t' , and newly entered activities at the current time t that satisfy the precedence constraints. Therefore, if j is a rigorously eligible activity, the predecessors of j have been finished, and regular resources required for j must be satisfied at the current decision moment.

For example, if the set $E_t = \{3, 4, 5\}$, we first sort the elements based on a specific PR in E_t and $E_t = \{5, 4, 3\}$. There exists no conflict between regular resources when activities 5 and 4 are scheduled simultaneously.

Algorithm 1: A Specific Heuristic Algorithm for Time-space Schedule and Space Allocation

```

1 Initialize  $t \leftarrow 0, E_t \leftarrow \emptyset, RE_t \leftarrow \emptyset, TE_t \leftarrow \emptyset, C_t \leftarrow \{1\}, A_t \leftarrow \emptyset, PP^t \leftarrow \{(0, 0, 0)\}, RPP_t \leftarrow \emptyset, APP \leftarrow \emptyset, ST \leftarrow \emptyset, FT \leftarrow \emptyset, rrl_{kt} \leftarrow R_k(\tau = 1, 2, \dots, T, k = 1, 2, \dots, K)$ ;
   // Loop when some activities are unscheduled.
2 while  $|FT| < |N|$  do
3   if  $|FT| = |N| - 1$  then
4      $FT \leftarrow FT \cup \{j\}; f_j \leftarrow \max\{FT_j | \forall j \in N - \{J\}\}$ 
5   else
6     // Identify  $E_t$  and extract  $RE_t$  that satisfy regular resource constraints from  $E_t$ .
7      $E_t \leftarrow E_t \cup TE_t \cup \{i | \forall i \notin A_t \cup C_t, P(i) \in C_t\}; TE_t \leftarrow \emptyset$ ;
8     use a specific priority rule to sort activities in  $E_t$ ;
9     for  $i$  in  $E_t$  do
10      if  $r_{ik} \leq rrl_{kt}; \forall \tau \in [t+1, t+d_i]$  then
11         $RE_t \leftarrow RE_t \cup \{i\}; E_t \leftarrow E_t - \{i\}; rrl_{kt} \leftarrow rrl_{kt} - r_{ik}$ 
12      end
13    end
14    for  $j$  in  $RE_t$  do
15      // Judge whether spatial resource constraints are satisfied for activity  $j$  in  $RE_t$ .
16      for  $p$  in  $PP^t$  do
17        // Judge whether  $j$  can be scheduled when using point  $p$  in  $PP^t$ .
18        if  $v_j$  uses  $p$  and satisfies the non-overlapping constraint then
19          if  $v_j$  not exceed  $V$  then
20            activity  $j$  is scheduled,  $s_j \leftarrow t, f_j \leftarrow t + d_j, PP^t \leftarrow PP^t - \{p\}, PP^t \leftarrow PP^t \cup \{PP_{pt}^j, PP_{pt}^j, PP_{pt}^j\}$ ;
21             $RE_t \leftarrow RE_t - \{j\}, A_t \leftarrow A_t \cup \{j\}, APP \leftarrow APP \cup \{p\}, ST \leftarrow ST \cup \{s_j\}, FT \leftarrow FT \cup \{f_j\}$ .
22          else
23             $RPP_t \leftarrow RPP_t \cup \{p\}, PP^t \leftarrow PP^t - \{p\}$ 
24          end
25        else
26           $RPP_t \leftarrow RPP_t \cup \{p\}, PP^t \leftarrow PP^t - \{p\}$ 
27        end
28      end
29      if  $|PP^t| = 0$  and  $j \in RE_t$  then
30        break
31      end
32      if  $j \notin RE_t$  then
33        break
34      end
35    end
36    if  $|PP^t| = 0$  and  $j \in RE_t$  then
37       $m = 0$ ;
38      // When the spatial constraints are still not satisfied for  $j$  by traversing  $p$  in  $PP^t$ , and then we will
39      // traverse point  $p_1$  in  $RPP_t$ .
40      for  $p_1$  in  $RPP_t$  do
41         $m = m + 1$ ;
42        if  $v_j$  uses  $p_1$  and satisfies the non-overlapping constraint then
43          if  $v_j$  not exceed  $V$  then
44            activity  $j$  is scheduled,  $s_j \leftarrow t, f_j \leftarrow t + d_j$ , update  $RE_t, A_t, APP, ST, FT$  (similar to line 18)
45             $RPP_t \leftarrow RPP_t - \{p_1\}, PP^t \leftarrow PP^t \cup \{PP_{p_1}^j, PP_{p_1}^j, PP_{p_1}^j\}$ 
46          end
47        end
48        if  $j \notin RE_t$  then
49          break
50        end
51        if  $m = |RPP_t|$  then
52          activity  $j$  can not be scheduled,  $TE_t \leftarrow TE_t \cup \{j\}, RE_t \leftarrow RE_t - \{j\}, rrl_{kt} \leftarrow rrl_{kt} + r_{jk}$ 
53          break
54        end
55      end
56    end
57    if  $|RE_t| = 0$  then
58      break
59    end
60  end
61  // Update the decision time  $t$  and the related sets.
62   $t \leftarrow \min\{FT_j | \forall j \in A_t\}$ ;
63   $C_t \leftarrow C_t \cup \{j | \forall j \in A_t | FT_j = t\}, PP^t \leftarrow PP^t \cup \{APP_j | \forall j \in A_t | FT_j = t\}, rrl_{kt} \leftarrow rrl_{kt} + r_{jk} (\forall j \in A_t, FT_j = t), A_t \leftarrow A_t - \{j | \forall j \in A_t | FT_j = t\}$ ;
64  end
65 end

```

However, activity 3 cannot satisfy the regular resource constraint. Therefore, at the current time t , the set of rigorously eligible activities $RE_t = \{5, 4\}$.

Generally, when scheduling some activities of a project, we rank activities in the set RE_t according to a specific PR, and the activity with the maximum or minimum priority value will be scheduled first. Different schedule sequences will affect our designed algorithm's performance, so it is an important issue to explore more effective PRs. Our new 3D-sRCPSP is an extension and improvement of the traditional RCPSP. Resources in the traditional RCPSP mainly refer to regular resources, which are represented only by a numerical value, without considering shape, volume and placement location. However, spatial resources significantly differ from regular resources in terms of capacity measurement, usage pattern and allocation mechanism. As a result, we cannot only adopt the PRs from the traditional RCPSP but also explore new PRs that are compatible with the characteristics of our 3D-sRCPSP. Firstly, our 3D-sRCPSP has the typical time-space dynamic characteristic, it continuously schedules activities along time axis and dynamically allocates 3D space for activities. Obviously, time and 3D space are crucial components of the 3D-sRCPSP. In addition, the space occupied by activities is mainly determined by their required regular resources. As activities start and finish, regular resources and spatial resources occupied by activities are dynamically occupied and released. Undoubtedly, regular resources and spatial resources have an interactive relationship. Therefore, time, regular resources and spatial

resources are important elements in the 3D-sRCPSP, and we can explore new forms of PRs based on them.

Besides the r-PRs shown in Table 3, these new forms of PRs are presented in Table 6. Our main motivation for selecting six composite index (CI) PRs comes from the study of Dixit et al. (2018) for the 2D-sRCPSP in shipbuilding projects, which is based on different combinations of the 2D spatial resource criticality index (SRCI-2D) and the time criticality index (TCI). Then summation, subtraction, multiplication and division operators are used, respectively. When two kinds of PRs are combined, the subtraction and the division can be used in the normal or reverse form. Therefore, CI-a, CI-b, CI-c, CI-d, CI-e and CI-f are obtained. Specifically, our SRCI is designed based on the 3D spatial resource. As a result, based on the selected PRs and operators, we can obtain six CI rules. In addition, TCI is the time criticality index, which is based on the total floats (TF) of activities. The TF indicates the maximum time that the start time of an activity can be delayed without affecting the project makespan, and it is calculated as the difference between the latest and the earliest start time of an activity. The maximum TF of all activities is denoted by $\text{Max}\{TF_{j \in N}\}$, then the difference between $\text{Max}\{TF_{j \in N}\}$ and TF_j of activity j is divided by $\text{Max}\{TF_{j \in N}\}$, we can obtain the TCI_j , where $TCI_j = (\text{Max}\{TF_{j \in N}\} - TF_j) / \text{Max}\{TF_{j \in N}\}$. However, the study of Dixit et al. (2018) focuses on the 2D-sRCPSP with 2D spatial resources, and their study does not consider the interactions between regular resources and spatial resources. Moreover, the characteristic of the 3D-sRCPSP is the interactive relations between time and 3D space, as well as between

Table 6
11 PRs based on the features of 3D spatial resources (s-PRs).

s-PR	Formula	Extreme
Spatial resource criticality index (SRCI)	$(l_j \times w_j \times h_j)/(L \times W \times H)$	Max
Composite index a (CI-a)	$TCI \times SRCI$	Max
Composite index b (CI-b)	$TCI + SRCI$	Max
Composite index c (CI-c)	$TCI - SRCI$	Max
Composite index d (CI-d)	$SRCI - TCI$	Max
Composite index e (CI-e)	$TCI / SRCI$	Max
Composite index f (CI-f)	$SRCI / TCI$	Max
Integrative combination (IC)	$w_1 \times TCI + w_2 \times RRCI + (1 - w_1 - w_2) \times SRCI (0 \leq w_1, w_2 \leq 1)$	Max
Greatest length (GL)	l_j	Max
Greatest width (GW)	w_j	Max
Greatest height (GH)	h_j	Max

regular resources and 3D spatial resources. Therefore, we introduce the integrative combination (IC) rule by combining the regular resource criticality index (RRCI), SRCI (3D) and TCI.

In addition, Crainic et al. (2008) used volume-first and height-first rules for the 3D-BPP, which produced better experimental results. Therefore, in this study, we not only consider volume-first and height-first, but also try to explore length-first and width-first rules. The volume-first rule refers to the larger volume of an activity, the greater priority value, we can transfer it to SRCI by normalization. Therefore, we only keep SRCI rule for the purpose of consistency. Greatest length (GL), greatest width (GW) and greatest height (GH) rules use the single dimension of length, width and height, respectively. One reason for choosing the single-dimensional size is motivated by the research on the 3D-BPP described above. Another reason is that less fragmented space is created when cubes are prioritized for placement in accordance with their sizes, which results in more efficient space utilization, because efficient use of space can serve to shorten a project makespan.

4.3. Dynamic generation strategy for placement points (PPDGS)

New placement points are dynamically generated in two scenarios. One is when activity j is scheduled. That is to say, when the required cuboid of j is allocated and occupied (just like a small cuboid v_j is placed into the total space V), it will generate three new placement points. The other is that activity j (v_j) is removed from V because j is finished. In the latter situation, the reference point (rp) of v_j can be transformed into one new placement point.

Scenario 1: The coordinates of rp for v_j are (x_{jt}, y_{jt}, z_{jt}) , and the placement points generated by v_j in X, Y and Z directions are PP'_{jx} (the right-back-below vertex of v_j), PP'_{jy} (the left-front-below vertex of v_j) and PP'_{jz} (the left-back-up vertex of v_j), respectively. Obviously, $PP'_{jx} = (x_{jt} + l_j, y_{jt}, z_{jt})$, $PP'_{jy} = (x_{jt}, y_{jt} + w_j, z_{jt})$ and $PP'_{jz} = (x_{jt}, y_{jt}, z_{jt} + h_j)$. Then, these generated placement points will be placed in the placement point set PP' , which is shown in lines 16–18 of Algorithm 1.

We take Fig. 6 as an example to illustrate the first scenario. The sizes (length, width and height) of v_i and v_j are (3,3,4) and (5,3,3), respectively. We consider point (0,0,0) as the rp of v_i . When activity i is executed in V (just like v_i is placed into V), three placement points are generated along X, Y and Z directions, which are $PP'_{ix} = (3,0,0)$, $PP'_{iy} = (0,3,0)$ and $PP'_{iz} = (0,0,4)$. At the same time, the used point (0,0,0) is removed from PP' . Therefore, $PP' = \{(3,0,0), (0,3,0), (0,0,4)\}$. If v_j of the next scheduled activity j uses PP'_{iz} (of v_i) as its rp , then again three placement points $PP'_{jx} = (5,0,4)$, $PP'_{jy} = (0,3,4)$ and $PP'_{jz} = (0,0,7)$ are created. PP'_{iz} is also removed from PP' . Therefore, the set PP' is dynamically updated along the progress of space allocations.

Scenario 2: During scheduling the activities of a project, at decision time t , the activity with the highest priority value in RE_t is first selected for scheduling and allocating space. If an available placement point is found, then this activity is arranged to execute and is deleted from RE_t . Then, we continue to check the next activity in RE_t . If there exists an

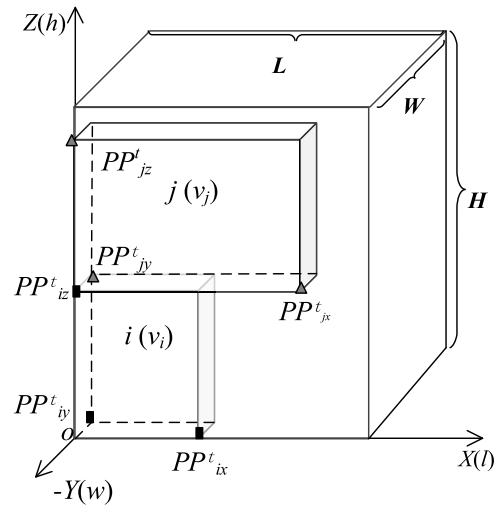


Fig. 6. Placement points defined by small cuboids v_i and v_j .

activity j in RE_t , we cannot find an available placement point for j . Activity j is removed from RE_t and added to TE_t . However, we still need to check the activities ranked after j in RE_t one by one. These activities can also be arranged if their spatial constraints are satisfied. If the set RE_t is empty, we must determine the next decision time t' ($t' > t$). When activity j' is finished at time t' , its occupied cuboid will be freed and its rp is idled. Therefore, this rp can be transferred to a new placement point and added to PP' , this is scenario 2 about the generation of placement points, which is shown in lines 51–53 and 55–56 of Algorithm 1. The set of E_t and RE_t is updated along a project's progress and the above procedure is repeated. Therefore, placement points in the set PP' are dynamically updated.

4.4. Dynamic search strategy for placement points (PPDSS)

Our PPDSS will search placement points in PP' one by one for activity j in RE_t . This search process is terminated when either of the two conditions is encountered for activity j : one is when activity j is scheduled; the other is when activity j is not scheduled, but the set PP' is empty. Our PPDSS corresponds to lines 14–31 of Algorithm 1.

To be more specific, we first select activity j with the highest priority values from RE_t , then start from the first placement point in PP' and check each placement point one by one to judge whether j can be scheduled. Placement point p can be regarded as a feasible location for activity j if two kinds of spatial constraints are fulfilled. The first type is that v_j does not overlap with ongoing cubes in V , which has been elaborated in detail in Section 3.3.1. The second type is $PP'[p][1] + l_j \leq L$, $PP'[p][2] + w_j \leq W$ and $PP'[p][3] + h_j \leq H$. Therefore, activity j can be scheduled. At the same time, we will obtain

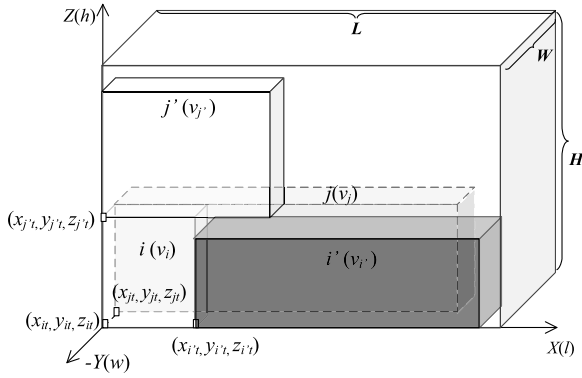


Fig. 7. 3D space of ongoing activities at time t .

the start time s_j and finish time f_j of j , and add s_j and f_j to ST and FT , respectively. p is also deleted from PP^t and three placement points generated by j are added to PP^t . Moreover, j is removed from RE_t and added to A_t . More importantly, we store the reference point information of v_j in V , so we add p to the set APP . Therefore, when activity j is scheduled, the search of the placement points in PP^t is terminated for j . In addition, if activity j' in RE_t visits placement point p in PP^t , the spatial constraints are not satisfied when arranging j' . We will remove p from PP^t and add it to RPP_t , then search for other placement points that rank behind p . When the set PP^t is empty and activity j' is still not scheduled, we also terminate the search of the elements in PP^t for activity j' .

Fig. 7 is used as an example to illustrate this strategy. At the current time t , the set $PP^t = \{(0,0,0)\}$, $RE_t = \{i, j, i', j'\}$, the space sizes in three dimensions for each activity are $i(3,3,4)$, $j(7,3,4)$, $i'(5,3,3)$ and $j'(4,3,3)$, respectively. The length, width and height of V are 9, 6 and 8, respectively. Firstly, if activity i has the highest priority, so we will first assign a placement point for i . Then, we check the placement point in PP^t one by one for activity i . The first placement point $(0,0,0)$ can be considered an available one for activity i . As a result, new placement points are generated, which are $PP_{ix}^t = (3,0,0)$, $PP_{iy}^t = (0,3,0)$ and $PP_{iz}^t = (0,0,4)$. At this time, $PP^t = \{(3,0,0), (0,3,0), (0,0,4)\}$, $RE_t = \{j, i', j'\}$. Secondly, we will look for a feasible placement point for the next activity j . If the placement point PP_{ix}^t is allocated to j , it will exceed the boundary of V . We add PP_{ix}^t to the set RPP_t , where $RPP_t = \{(3,0,0)\}$. At this time, $PP^t = \{(0,3,0), (0,0,4)\}$. Then, we search other placement points in PP^t and find the point $(0,3,0)$ that can be assigned to activity j . Three placement points PP_{jx}^t , PP_{jy}^t and PP_{jz}^t are generated simultaneously, where $PP^t = \{(0,0,4), (7,3,0), (0,6,0), (0,3,4)\}$. Suppose all placement points have been searched and no one is available for activity j , placement points are all deleted from PP^t and PP^t becomes the empty set, so j has to be delayed to the next decision moment if we do not consider the RPPSAS. The other activities (i' and j') in RE_t will also judge whether spatial constraints are satisfied according to the RPPSAS.

4.5. Secondary allocation strategy for residual placement points (RPPSAS)

Before elaborating on the secondary allocation strategy for residual placement points, we need to clarify what a residual placement point is and what kind of points can be included in the residual placement point set (RPP_t).

When placement point p is unavailable for activity j in RE_t , it is considered as a residual placement point. In other words, when v_j uses p , it will overlap with other ongoing cuboids in V or exceed the boundary of V . Then, p in PP^t is converted to a residual placement point and added to the set RPP_t . All generated residual placement

points are placed in RPP_t . The set RPP_t is updated dynamically along the time axis.

Although one residual placement point is not available for activity j in RE_t , it may be available for another activity j' in RE_t . Therefore, it is necessary to reallocate these residual placement points. Our RPPSAS corresponds to lines 32–50 of Algorithm 1.

If activity j searches all placement points in PP^t and still cannot find an available one, we need to continuously allocate residual placement points that have been added to RPP_t because other activities in RE_t are not able to use them. For activity $j \in RE_t$, there are also two conditions to terminate the search for the residual placement points in RPP_t . One is when activity j is scheduled by using the point in RPP_t , the allocation for residual placement points is ended. The other is when activity j still cannot be scheduled by traversing all elements in RPP_t ($m = |RPP_t|$), which is shown in line 45 of Algorithm 1, this is the first difference from the PPDSS.

In order to find a feasible residual placement point p_1 for activity j , we check each residual placement point in RPP_t one by one from the first element. If the spatial constraints are satisfied, p_1 can be considered as the reference point for j (the location of v_j in V). Therefore, activity j can be scheduled, and the update method of sets RE_t , A_t , APP , ST and FT is similar to that for the PPDSS. However, p_1 must be removed from RPP_t and three new placement points are added to PP^t , this is the second difference from the PPDSS. Otherwise, we will search other residual placement points in RPP_t for activity j . Activity j will not be scheduled when all points in RPP_t are unavailable for j , and j has to be delayed to the next decision moment t' . As a result, we will remove j from RE_t and add j to the temporary set TE_t , and $rrl_{k\tau}$ is also updated, this is the third difference from the PPDSS.

At the current time t , if spatial constraints are not satisfied for activity j after searching for all elements in PP^t and RPP_t , we add activity j to the temporary set TE_t . The main reason is to prepare for determining sets $E_{t'}$ and $RE_{t'}$ at the next decision time t' . At the moment t' , $E_{t'}$ is composed of remaining activities in E_t at time t , activities in TE_t that spatial resources cannot be satisfied at time t , and new activities that satisfy the precedence constraint based on the completed activities at time t' .

5. Computational experiments

In this section, extensive numerical experiments are implemented to validate our 3D-sRCSP model and customized heuristic algorithm. The datasets for the 3D-sRCSP are constructed in Section 5.1. Section 5.2 devises the adjustment measures when time-space conflicts are encountered. The test indicators for numerical experiments are designed in Section 5.3. Section 5.4 tests the effectiveness of different PRs. Section 5.5 demonstrates the effectiveness of the model and the performance of our customized heuristic algorithm. Section 5.6 analyzes the number of time-space conflicts under different strategies. Section 5.7 presents comparative analyses between heuristic solutions and optimal solutions. The extension research on the adaptive selection of PRs based on project characteristics is shown in Section 5.8. The computational experiments are conducted on a PC with an Intel(R)Core(TM)i7-8700CPU, 8 GB RAM, and Windows 10 operating system, and the algorithms are programmed in Python.

5.1. Generation of numerical instances

The 3D-sRCSP is a new problem proposed by this paper, so there are no available instances for our 3D-sRCSP in the widely used PSPLIB. To gain two different types of instances (R-instances and S-instances), the benchmark instances of the traditional RCSP in PSPLIB are modified to configure parameters for spatial resources, which are deliberately explained as follows:

(1) The 3D space demand data for each activity is obtained through its regular resource requirements. Therefore, the sum of regular resource requirements for each activity is decomposed into the product of three integers as the sizes in three dimensions (l_j , w_j and h_j) of its occupied cuboid. The three maximal values in three dimensions of the cuboids occupied by all activities are selected as the length (L), the width (W) and the height (H) of a total construction site (V). As a result, we can obtain R-J30, R-J60, R-J90 and R-J120 instances for the 3D-sRCPSP, they are regarded as R-instances with a total of 2040 instances.

(2) The 3D space demand data for each activity is generated randomly. The sizes (L , W and H) of V on three dimensions are all determined by the number of activities. To guarantee that each cuboid can be placed in the V horizontally or vertically, and cannot exceed the boundary of V , the length (l_j), the width (w_j) and the height (h_j) of each cuboid are randomly generated based on the sizes of V . Therefore, we can obtain the S-J30, S-J60, S-J90 and S-J120 instances for the 3D-sRCPSP, they are considered S-instances with a total of 2040 instances.

For each instance, the optimal solution and its shortest makespan from the traditional RCPSP are regarded as the relaxed baseline schedule scheme ($RBSS$) and the lower bound, respectively, which are used to validate the model and the heuristics. In addition, in order to compare the solutions obtained by our heuristic algorithm with that from Gurobi, we adopt instance generator RanGen to produce 8 groups of small-scale instances with a total of 10 non-dummy activities based on order strength (OS) and resource strength (RS) (OS=0.3, 0.7 and RS=0.25, 0.5, 0.75, 1) (Demeulemeester, Vanhoucke, and Herroelen, 2003). There are eight sets of instances, with a total of 40 instances. The sizes of the 3D cuboid for each activity are determined according to the generation method of R-instances described above. The generated instances are named R-J10. Similarly, we can also produce the instances with 15 activities and consider them as R-J15. For R-J10 and R-J15, there are a total of 80 instances.

5.2. Reactive adjustment measures

In practical scenarios, if 3D spatial constraints are ignored in the planning stage for a class of projects with limited construction sites, the $RBSS$ may not be feasible. When the project is implemented according to its $RBSS$, some activities will not be executed smoothly due to the occurrence of time-space conflicts. Then the $RBSS$ has to be temporarily adjusted several times. In other words, project managers can only passively take remedial measures to deal with these conflicts that actually occur in construction sites. There is no time to change construction methods and designs but postpone some activities, which ultimately leads to delays in the project makespan. Therefore, we refer to the reactive project scheduling theory for the RCPSP (Deblaere, Demeulemeester, and Herroelen, 2011), and design our reactive adjustment measure for the $RBSS$ that ignores time-space conflicts. Our adjustment measure relies on a reactive procedure to adapt the schedule to new information and restore its feasibility. When time-space conflicts are encountered, we design the reactive adjustment measures, but the start time of each activity cannot be earlier than its planned start time.

We take Fig. 8 as an example to illustrate the reactive adjustment measure. In Fig. 8a, activities 1 and 6 are the start and the end activities (dummy), respectively. The durations and regular resource requirements for all non-dummy activities are 1, and the availability of regular resources is 2. Firstly, the $RBSS$ is shown in Fig. 8b. However, this $RBSS$ becomes infeasible due to time-space conflicts. At the initial decision time ($t=0$), if there exist conflicts between activities 2 and 3, and activity 2 has a higher priority than activity 3 according to a specific PR. Therefore, we need to postpone activity 3 to the next decision moment ($t=1$), as shown in Fig. 8c. However, the sequence of the local scheduling scheme (activities 4 and 5) behind activity 3 is not

changed but only postponed to the next decision time ($t=2$). Finally, time-space conflicts are eliminated and the actual project makespan is 3.

5.3. Performance indicators in numerical experiments

The performance indicators in our numerical experiments are adopted to validate the model and the heuristics of the 3D-sRCPSP. DM_1 , DM_2 and DM_3 are designed for testing R-instances and S-instances. DM_4 is designed for testing the small-scale instances R-J10 and R-J15. For a numerical instance in R-instances and S-instances, D_d is used to denote its actual project makespan, which is obtained based on the reactive measure by adjusting the $RBSS$. D_s refers to the project makespan of the 3D-sRCPSP obtained by our designed PRs heuristics. That is to say, time-space conflicts are resolved in the project planning stage for the 3D-sRCPSP. The shortest project makespan of the traditional RCPSP is obtained from the PSPLIB library as D_{opt} . Therefore, two deviations between D_s and D_{opt} , D_d and D_{opt} are considered to design performance indicators DM_1 and DM_2 , respectively. Moreover, DM_3 refers to the deviation between D_d and D_s . For a small-scale instance in R-J10 and R-J15, the optimal (minimum) project makespan D_{opt1} of the 3D-sRCPSP can be achieved by Gurobi, which serves as the lower bound of D_s for designing the deviation DM_4 between D_s and D_{opt1} . Based on the description above, the following experimental test metrics are shown in Eqs. (31)–(34) for each instance. As a result, for the total set of instances, the average DM_1 , DM_2 , DM_3 and DM_4 are obtained to verify the effectiveness of the model and algorithm.

$$DM_1 = \frac{|D_s - D_{opt}|}{D_{opt}} \times 100\% \quad (31)$$

$$DM_2 = \frac{(D_d - D_{opt})}{D_{opt}} \times 100\% \quad (32)$$

$$DM_3 = \frac{(D_d - D_s)}{D_d} \times 100\% \quad (33)$$

$$DM_4 = \frac{(D_s - D_{opt1})}{D_{opt1}} \times 100\% \quad (34)$$

5.4. Comparative analyses of the performance on different PRs based heuristic algorithm

5.4.1. Comparisons of performance on (r-PR)-PSGS-SAS

Our heuristic algorithm consists of three parts: r-PR, PSGS and SAS. Therefore, we label the r-PR based heuristic algorithm as (r-PR)-PSGS-SAS and compare the performance of (r-PR)-PSGS-SAS under R-instances and S-instances. The average DM_1 of (r-PR)-PSGS-SAS under different scale instances is shown in Fig. 9. The purple line represents the average performance of (r-PR)-PSGS-SAS on all scale instances. From Fig. 9a, we observe that the overall trend is that as the sizes of instances improve, the average of DM_1 in S-instances continues to decline. However, when the sizes of instances grow to 120 activities (S-J120), the average of DM_1 is slightly higher than that for 90 activities (S-J90) but significantly lower than that for 30 activities (S-J30) and 60 activities (S-J60). This changing trend indicates that our designed heuristic algorithm works better in solving medium and large scale instances. Obviously, the average of DM_1 (15.05%) of EFTD-PSGS-SAS is the smallest, but the average of DM_1 of RAD-PSGS-SAS is the largest. Although the average of DM_1 fluctuation of all instances is not significant, the fluctuation of the average of DM_1 is still relatively large under different scale instances, especially for S-J30 instances.

For R-instances in Fig. 9b, the difference of average DM_1 between small-scale and medium-scale instances is very small when these instances are solved by our heuristic algorithm. However, the average of DM_1 increases abruptly when solving large-scale instances. The sizes of an activity's 3D spatial resource for R-instances are constructed by the sum of demanding amounts of regular resources. When the size of

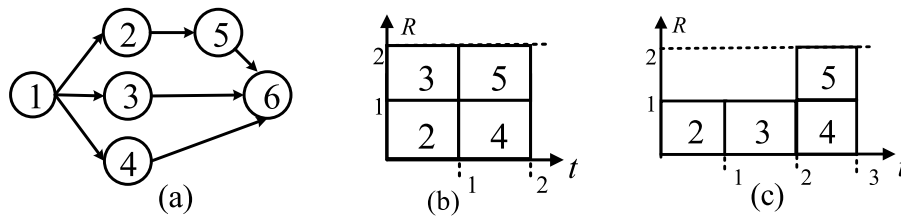


Fig. 8. An example of the reactive adjustment measure.

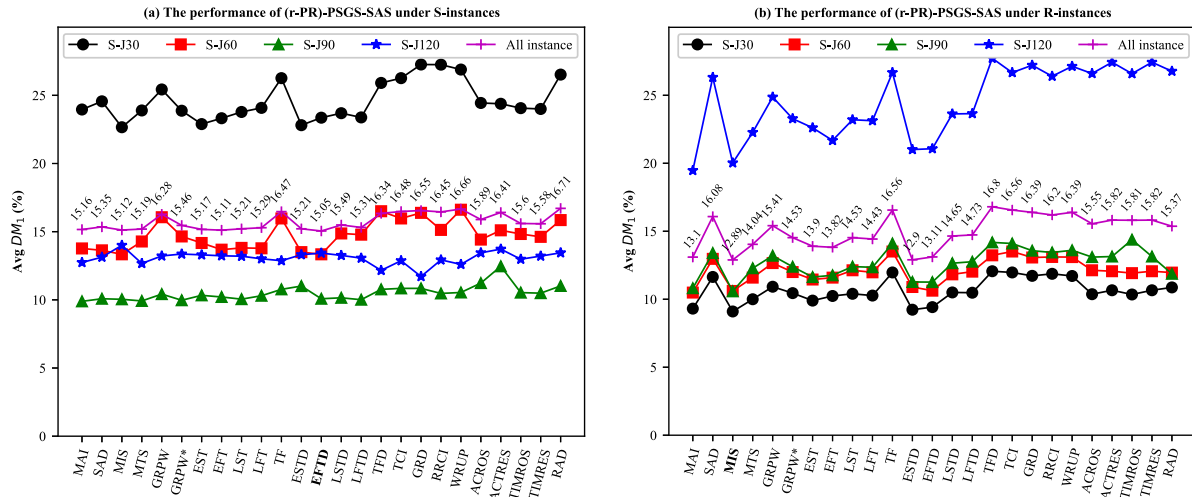


Fig. 9. Performance of r-PR based heuristic algorithm on S-instances and R-instances.

instances increases, the difficulty of space allocation also greatly climbs. Comparing Fig. 9a and Fig. 9b, the average of DM_1 of RAD-PSGS-SAS is the largest under S-instances. However, the average of DM_1 of RAD-PSGS-SAS is in a moderate position under R-instances, some PRs are worse than the random rule.

5.4.2. Comparisons of performance on (s-PR)-PSGS-SAS

The average of DM_1 for S-instances and R-instances under 11 s-PRs are shown in Fig. 10. In Fig. 10, the colors in the bar represent different scale instances. The horizontal axis represents different (s-PR)-PSGS-SAS. For each s-PR, there are two bars: the left represents the average DM_1 of the (s-PR)-PSGS-SAS under different scale instances, and the right represents the average DM_1 of the (s-PR)-PSGS-SAS by considering all instances.

It is immediately clear from Fig. 10 that the average of DM_1 for S-instances and R-instances indicates better performance from IC-PSGS-SAS and SRCI-PSGS-SAS, and IC-PSGS-SAS is always the best. The performance improvement of IC-PSGS-SAS is not significant compared with SRCI-PSGS-SAS, because SRCI is also based on 3D spatial resources, which indirectly proves the effectiveness of considering 3D spatial resources in rules. Among these s-PRs, comparing Fig. 10 with Fig. 9, IC rule further improves the performance of our heuristic algorithm compared with the PRs selected from the traditional RCPSP, especially for R-instances. Because spatial resources in R-instances are obtained by decomposing regular resources, PRs are used to select the activities that can be prioritized to schedule from the set of eligible activities at each decision moment, and the IC rule combines time, regular resources and spatial resources. Moreover, we schedule the activities of a project along the time axis. There exists an interactive relation between regular and spatial resources in the 3D-sRCPSP.

The average of DM_1 under different weight combinations of w_1 and w_2 for IC rule is shown in Table 7, and its changes in trend are visualized in Fig. 11. Firstly, we clearly see that the better weight combination of w_1 and w_2 for IC rule is 0.3 and 0.3, respectively. In this

scenario, the average of DM_1 is the smallest. Moreover, the second-best combination of weight is when $w_1 = 0.4$ and $w_2 = 0.3$, followed closely by $w_1 = 0.3$ and $w_2 = 0.4$, as well as $w_1 = 0.4$ and $w_2 = 0.4$. In these scenarios, the weights of TCI, RCI and SRCI are close to equal, that is, considering TCI, RCI and SRCI in balance. Secondly, it can be observed from Table 7 that when the sum of w_1 and w_2 is equal to 1, the average of DM_1 is relatively large and is the value on the diagonal of Table 7. In this situation, only time and regular resource criticality are considered, and spatial resource criticality is not included. Thirdly, when w_1 and w_2 are all equal to 0, IC rule is converted to SRCI, and the average of DM_1 equals 12.654.

5.4.3. Statistical comparisons

According to the illustration in Sections 5.4.1 and 5.4.2, we selected six r-PRs and two s-PRs that exhibit better performance when solving different scale instances. In addition, RAD rule and poorly performing TFD rule are chosen as the benchmark for comparison. Since there is no direct conclusive evidence of the superiority of any PR over others for multiple types of instances, we conduct the Friedman test to rank these selected PRs, which is shown in Table 8. After getting relative ranks for the multiple types and different scales of 3D-sRCPSP instances, we can obtain the mean ranks for each scale instance under our selected ten PRs. The last row (total) indicates the sum of mean ranks. The smaller value obtained by the Friedman test, the higher ranking of the PR.

In general, we can observe that IC, SRCI and MIS rules are in the top three positions under different scales and different types of instances from Table 8. MIS outperforms other r-PRs under small-scale instances (J30) and medium-scale instances (J60), but ESTD has better performance under J30, J90 and J120 instances. Our computational results on different scale instances show that the best PRs for the traditional RCPSP not perform best for the 3D-sRCPSP under time-space conflict environments. Moreover, IC performs well among all scales and all types of 3D-sRCPSP instances, which combines time, regular resources and spatial resources. Therefore, it more closely matches the

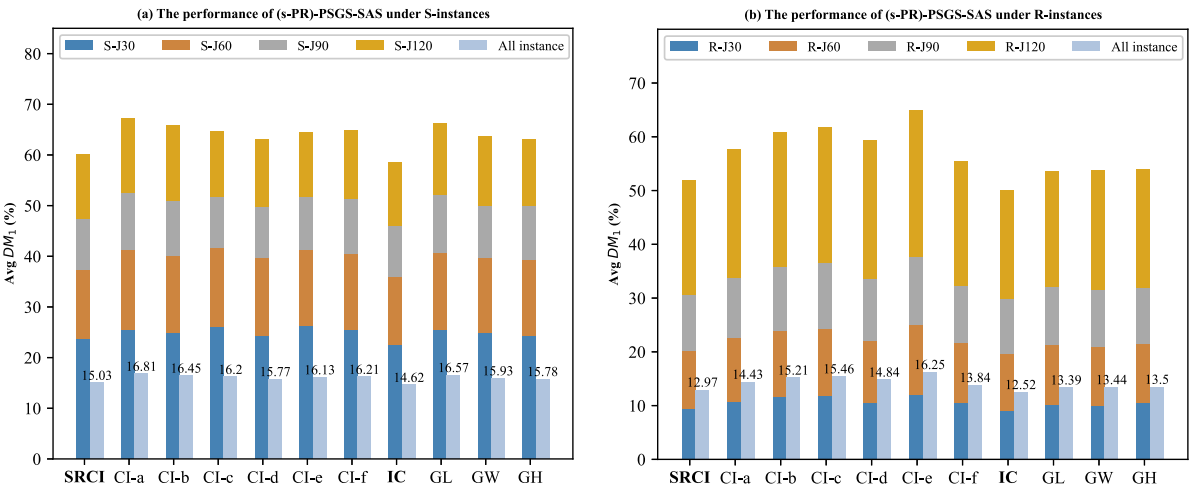


Fig. 10. Performance of s-PR based heuristic algorithm on S-instances and R-instances.

Table 7
The average of DM_1 under different weight combinations of w_1 and w_2 for the IC rule.

$w_1 \backslash w_2$	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
0	12.654	12.643	12.641	12.637	12.636	14.923	12.645	12.723	12.777	12.843	12.873
0.1	12.644	12.627	12.602	12.602	12.612	12.621	12.634	12.654	12.734	13.651	–
0.2	12.631	12.617	12.598	12.577	12.586	12.646	12.687	12.743	12.987	–	–
0.3	12.630	12.614	12.588	12.564	12.567	13.543	13.732	12.843	–	–	–
0.4	12.628	12.607	12.574	12.568	12.571	14.024	13.876	–	–	–	–
0.5	12.717	12.677	12.649	12.632	12.637	14.534	–	–	–	–	–
0.6	12.779	12.714	12.698	12.675	12.954	–	–	–	–	–	–
0.7	12.845	12.797	12.731	12.894	–	–	–	–	–	–	–
0.8	12.894	12.865	13.776	–	–	–	–	–	–	–	–
0.9	12.922	13.907	–	–	–	–	–	–	–	–	–
1	12.927	–	–	–	–	–	–	–	–	–	–

Table 8
Relative ranks for S-instances and R-instances.

Instance type	MIS	MAI	ESTD	EST	EFTD	EFT	IC	SRCI	RAD	TFD
S-J30	4.52	4.71	4.54	4.57	4.57	4.56	4.50	4.58	4.92	4.81
R-J30	3.56	3.51	3.53	3.55	3.53	3.57	3.56	3.59	4.01	4.62
Mean rank	4.04	4.11	4.04	4.06	4.05	4.07	4.03	4.09	4.47	4.72
S-J60	3.66	3.80	3.72	3.88	3.71	3.78	3.65	3.72	4.31	4.16
R-J60	3.68	3.62	3.73	3.75	3.70	3.79	3.62	3.61	4.01	4.22
Mean rank	3.67	3.71	3.73	3.82	3.71	3.79	3.64	3.67	4.16	4.19
S-J90	3.36	3.35	3.29	3.30	3.32	3.31	3.34	3.35	3.71	3.64
R-J90	3.35	3.37	3.39	3.42	3.39	3.43	3.34	3.35	3.65	4.35
Mean rank	3.36	3.36	3.34	3.36	3.36	3.37	3.34	3.35	3.68	4.00
S-J120	3.73	3.67	3.68	3.69	3.70	3.67	3.63	3.64	3.90	3.85
R-J120	4.55	4.58	4.55	4.59	4.56	4.58	4.53	4.53	5.22	5.36
Mean rank	4.14	4.13	4.12	4.14	4.13	4.13	4.08	4.09	4.56	4.61
Total	15.21	15.31	15.22	15.38	15.24	15.35	15.09	15.19	16.87	17.51

characteristics of the 3D-sRCPSP. In addition, by considering different types of instances together, the average performance of TFD rule is relatively poor and RAD rule is some distance away from our choice of a good one.

5.5. Analyses of model effectiveness and algorithm performance for 3D-sRCPSP

5.5.1. Analyses of model effectiveness

In order to validate the effectiveness of 3D spatial resource constraints in the 3D-sRCPSP model, we conduct numerical experiments through our designed heuristic algorithm based on the best PRs. Fig. 12 reports the average DM_1 , DM_2 and DM_3 with different NC (network complexity) and different scale instances (S-J30 and S-J120). Here,

NC depicts the average number of non-redundant arcs at each node that includes the dummy activities (Sprecher and Kolisch,1996). As observed in Fig. 12, the average of DM_1 is much smaller than that of DM_2 for those instances with different NC and sizes, which denotes that the average project makespan obtained from the 3D-sRCPSP is shorter than the average of actual makespan by adjusting the $RBSS$. Therefore, it is very necessary and important to eliminate time-space conflicts in the planning stage for these complex projects with limited construction sites. In addition, the minimum average of DM_3 is 22.23% under different NC in S-J30 instances, indicating that the 3D-sRCPSP can reduce the average project makespan by at least 22.23%. Better results can also be obtained in S-J120 instances. Therefore, it can be proved that our model for the 3D-sRCPSP is effective.

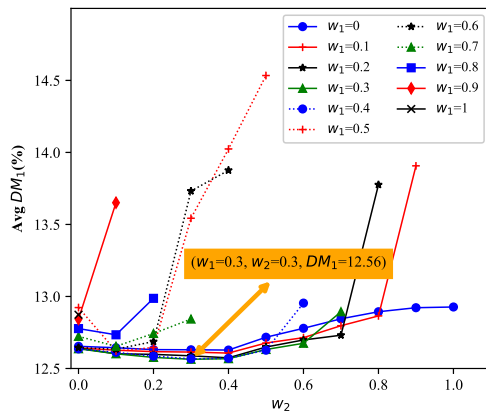


Fig. 11. The trends of average DM_1 under different weight combinations of w_1 and w_2 for IC rule.

5.5.2. Analyses of heuristic algorithm performance

Three space allocation strategies are the most important elements in our designed heuristic algorithm. In order to compare with our algorithm, we list two reference algorithms. We only add the first reference strategy to the PR heuristic algorithm for the traditional RCPS, which forms Algorithm a. Here, the first reference strategy (strategy 1) is to search for placement points in PP' for any activity j in RE , and postpone activity j whenever an unsuitable placement point is encountered. Then we incorporate the second reference strategy into the PR heuristic algorithm for the traditional RCPS and regard it as Algorithm b. The second reference strategy (strategy 2) is to search for placement points for activity j , when we find an unsuitable placement point, we will continue to search for other placement points in PP' . If we still cannot find an available point after searching for all placement points in PP' , activity j will be postponed. Algorithm c is our proposed heuristic algorithm, the space allocation strategies (strategy 3) in Algorithm c include PPDGS, PPDSS and RPPSAS. In Algorithm c, we will first search placement points for activity j . When all placement points in PP' are searched, but no available points are found, we continue to look for residual placement points in RPP_i until all points are traversed. We will postpone activity j when a feasible placement point is still not found. When we allocate the space to an activity, the three strategies (1, 2 and 3) differ in the number of searching for placement points (termed as the breadth). Namely, the breadth sequentially increases with strategies 1, 2 and 3. Fig. 13 reports the average DM_1 for different algorithms under S-J30 and S-J120 instances with different NC.

For S-J30 and S-J120 instances, the same trend we observed is that the average deviation obtained by Algorithms b and c is much lower than that from Algorithm a. In Algorithm a, if the current placement point cannot be available for the selected activity, this activity is directly postponed to the next decision moment without searching for other placement points in PP' and RPP_i . Therefore, Algorithm a only has one basic space allocation strategy, and the average of DM_1 is slightly inferior to that from Algorithms b and c. In addition, we also find that the average of DM_1 obtained by Algorithm c does not change much as NC of instances increases, showing that the designed Algorithm c in this paper displays good adaptability and stability for solving instances with different NC. As the size of 3D-sRCPS instances increases, the average of DM_1 obtained by Algorithm c decreases. This indicates that our designed heuristics can effectively deal with time-space conflicts for large-scale 3D-sRCPS instances.

5.6. Comparative analyses for the number of time-space conflicts

Time-space conflicts usually occur when scheduling the activities of a project according to its $RBSS$. Obviously, different space allocation

strategies combined with reactive measures must affect the number of time-space conflicts. At each decision moment, the spatial resources required by some eligible activities may not be satisfied, because these activities will compete with those being implemented activities for the limited construction site (space conflicts at a decision time). The amount of total activities involved in space conflicts at all decision moments is regarded as the number of time-space conflicts. Then, the number of time-space conflicts is compared among multiple types of instances with different NC and sizes. S-J30 and S-J120 instances are selected to execute the numerical experiments. Strategy 1X and strategy 2X are strategy 1 and strategy 2 in Section 5.5.2, respectively. Strategy 3X is our designed three space allocation strategies (PPDGS, PPDSS and RPPSAS). Therefore, the reactive scheduling measures with strategy 1X, strategy 2X and strategy 3X are used to resolve time-space conflicts, respectively. The numerical experiment results are shown in Fig. 14.

It can be immediately seen from Fig. 14 that time-space conflicts can frequently occur among activities when the $RBSS$ is executed. However, if we adjust the $RBSS$ during the execution of activities, the number of time-space conflicts based on strategies 2X and 3X will be significantly lower than that based on strategy 1X. This indicates that different space allocation strategies have a significant impact on the number of time-space conflicts when we adjust the $RBSS$. Moreover, with the increase of NC, the number of time-space conflicts will decrease for different space allocation strategies, but strategy 3X has the greatest reduction. In addition, comparing Fig. 14a with Fig. 14b, the number of time-space conflicts increases notably when the scale of the instances becomes larger, it demonstrates that large-scale projects are more prone to exist serious time-space conflicts.

5.7. The comparative analyses between heuristic solutions and optimal solutions

This subsection is devoted to comparing the PRs heuristics with the commercial optimization solver for solving the 3D-sRCPS. Since Gurobi can only obtain optimal solutions for those instances with very small sizes, we conduct experiments with R-J10 and R-J15 instances. The average computation time and the average deviation of project makespan for different scale instances are listed in Table 9. Column “INS” represents the names of different instances distinguished by R-J10 and R-J15. “OS” is the order strength, and “RS” is the resource strength. Columns “GT” and “HT” indicate running time in seconds using Gurobi and our heuristic algorithm, respectively. Column “ DM_4 ” is the makespan deviations between Gurobi (G) and our heuristics (H), which is just defined in Section 5.3. In addition, the “Avg” denotes the average value of a set of instances.

Firstly, we can get a detailed observation from Table 9 that with the increment of instance scale, Gurobi always takes longer time to solve the instances of the 3D-sRCPS than the heuristics, and this phenomenon is most apparent for smaller RS instances. The main reason is that the space required for an activity is determined by the regular resources it occupies. The regular resources and the spatial resources are tighter for smaller RS. Secondly, when OS becomes smaller, the average computation time is even longer, which is especially evident in Gurobi. Thirdly, in general, the average of DM_4 is all within 10%, and its value is 0 when RS equals 1. When OS is larger, the average deviation is smaller, indicating that our heuristic algorithm is better at solving instances with high OS.

Figs. A.1 and A.2 in the appendix visualize the comparisons of the makespan obtained by Gurobi and PR heuristics under different OS and RS, respectively. We find that the project makespan obtained by the heuristics or Gurobi gradually increases with the growth of OS. However, it decreases with the increase of RS. This is mainly due to the fact that the smaller RS value, the higher strength of resource constraints, and the higher OS value, the more precedence relation restrictions among activities.

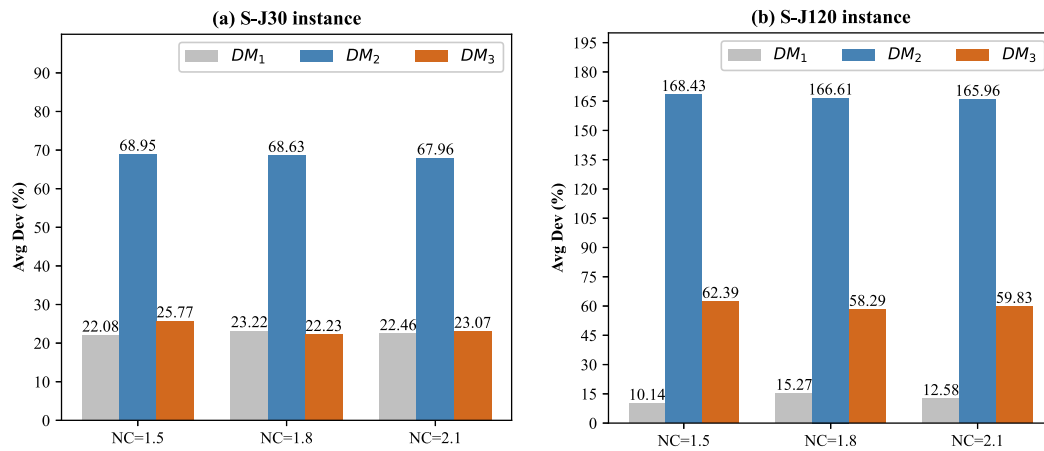


Fig. 12. Effectiveness of the model for S-J30 and S-J120 instances with different NC.

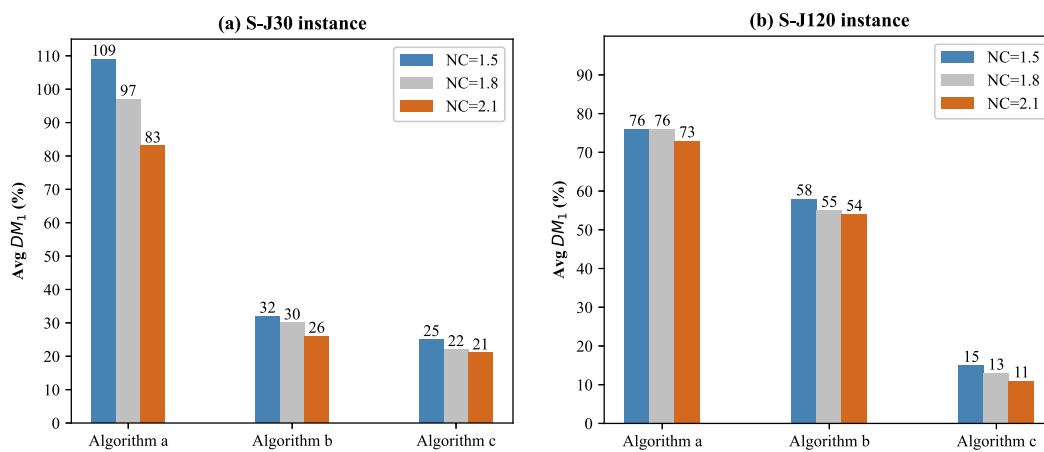


Fig. 13. Effectiveness of the heuristics for S-J30 and S-J120 instances with different NC.

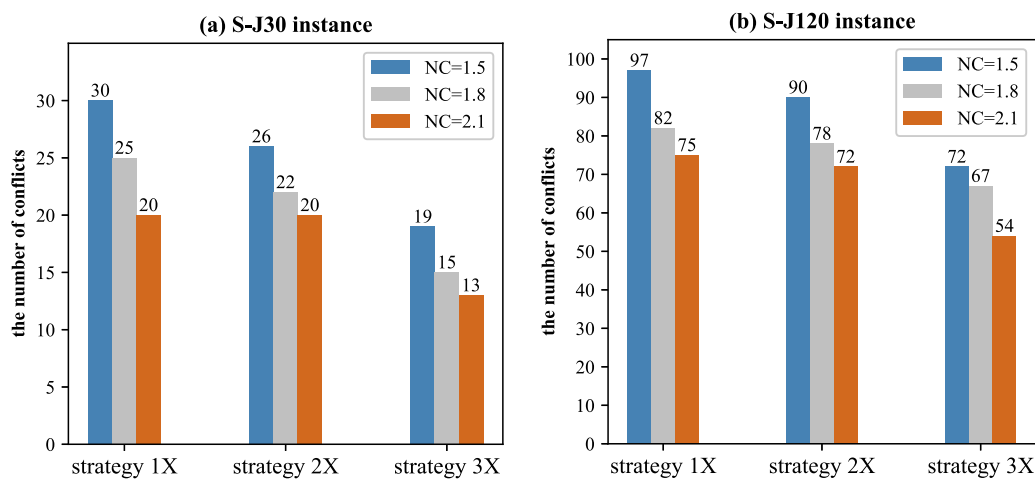


Fig. 14. The number of time-space conflicts under different strategies and NC.

In addition, Table A.1 in the appendix shows the experimental results for each instance of R-J15 and R-J10. Column “G” gives optimal solutions from Gurobi, and “H” is heuristic solutions obtained by our designed heuristic algorithm for each instance. The results show that most instances (63%) reach their optimal solutions by our heuristics and it has the average 0 of DM_4 , which is shown in bold in Table A.1 in the appendix. In particular, it is easier and quicker to find the optimal solutions for those instances with larger RS.

5.8. Extension: automatic selection of the best PR based on the machine learning technique

This section is an extension of the above work, which aims to adaptively select the best rule for a particular project instance with different characteristics, as we have found in numerical experiments that some PRs may not perform the best overall, but they exhibit better

Table 9
Comparisons of average performance between Gurobi and heuristic algorithm for R-J15 and R-J10 instances.

INS	OS	RS	Avg GT(s):Gurobi	Avg HT(s):Heuristics	Avg DM_4 (%)
R-J15	OS=0.3	RS=0.25	1236	24	10
		RS=0.5	209	14	9
		RS=0.75	53	15	5
		RS=1	24	12	0
	Avg	–	381	16	6
	OS=0.7	RS=0.25	480	20	1
		RS=0.5	75	8	4
		RS=0.75	40	7	6
		RS=1	8	2	0
	Avg	–	151	9	3
R-J10	OS=0.3	RS=0.25	107	20	8
		RS=0.5	23	5	6
		RS=0.75	12	6	5
		RS=1	2	2	0
	Avg	–	36	8	5
	OS=0.7	RS=0.25	13	4	0
		RS=0.5	10	4	3
		RS=0.75	9	3	1
		RS=1	1	2	0
	Avg	–	8	3	1

performance for a specific project instance. Therefore, we use a well-known classification decision tree method C4.5 in the machine learning technique to construct the decision tree model. We can automatically select the best PR based on the project features when the decision tree is constructed. C4.5 not only has fast classification speed, high accuracy and easy-to-understand classification rules, but also can handle continuous attributes. For more details on how the C4.5 method is used to construct the decision tree model, we can refer to Section 5.8.1. The performance evaluation of the generated decision tree model is shown in Section 5.8.2.

5.8.1. The creation of decision tree model

The decision tree is mainly composed of nodes and branches (edges), where the nodes include the root node, internal nodes and leaf nodes. The root node and internal nodes primarily represent the elements in the feature space, while leaf nodes mainly represent the elements in the label space. Branches represent the possible range of values for a certain feature. A path connecting from the root node of a tree through directed edges to a certain leaf node represents a classification rule. Therefore, in our decision tree, feature space and label space are two important components, and how to establish the decision tree model based on features and labels is also a key point. Then, we will introduce these one by one.

Feature space and label space. Some studies indicate that activity, network and resource (regular) indicators are the project feature parameter for the traditional RCPSP (Vanhoucke, Coelho, and Batselier, 2016; Vanhoucke and Coelho, 2018; Guo, Vanhoucke, and Coelho, 2023). Therefore, we use the activity indicator, network indicators and resource indicators as the elements in the feature space, as they are all important indicators for measuring the characteristics of a project. In addition, we use the measurement method of regular resource constrainedness to generate the spatial resource constrainedness indicator, the details can refer to the study of Demeulemeester et al. (2003). These project indicators are presented in Table 10, the range of indicator values for different scale instances is separated by slashes. It is obvious that the values of these indicators are continuous values, we need to discretize these continuous values when constructing a decision tree. Therefore, we use the binary method to discretize them and divide the dataset into two subsets based on a certain partition point on a certain feature. The specific steps are as follows: given the dataset D

and continuous attribute a , if a has n different values on D , we first sort these values in ascending order, denoted as $\{a^1, a^2, \dots, a^n\}$. Based on the partition point η , $\eta = \{\frac{a^i + a^{i+1}}{2} | 1 \leq i \leq n\}$, D can be divided into subsets D^- and D^+ , where D^- contains datasets with values on attribute a that are not greater than η , and D^+ includes datasets with values on attribute a that are greater than η . Then, we select the partition point η with the highest information gain to discretize a , the study of Guo, Vanhoucke, Coelho, and Luo (2021) gives the detailed calculation of information gain.

For each project instance, the heuristic algorithm under different PRs can obtain the corresponding project makespan. There always exists one or more PRs that perform best for a specific project instance. Therefore, we can determine whether the performance of a project instance is good or not under a certain PR, which can be used as one label. The performance (good or not) of a project instance under the 36 PRs can be considered as elements in the label space. Obviously, the number of features is 13 (indicator name in Table 10) and the number of labels is 36 in our study. Moreover, we need to construct one decision tree for each label. When the decision tree is constructed, the testing project instance can be predicted by tracing the path from the root node through branches to a leaf node. Then, we can determine whether the performance of the testing project instance under the corresponding label (one PR) reaches the best level or not.

Decision tree model. The goal of the decision tree model is to construct a tree structure, where the root node and internal nodes are composed of features, the values on the branches under one node represent the range values of the feature, and the leaf nodes represent the label (whether a rule is good (Y) or bad (N)). From the root node, branches to internal nodes and finally to leaf nodes, a complete decision tree is formed. Based on the decision model, we give a project's features to determine whether the better project performance can be achieved under the label of this model. Therefore, we can adaptively select the best PR for one project under the generated decision tree model. However, there exist many features in the feature space, it is crucial to choose which feature can be used as the root node, how to branch below the selected root node and when to stop branching to form the complete decision tree model.

The steps of constructing the decision tree model are as follows: Firstly, we need to prepare training datasets that know the feature space and label space of each project instance. Secondly, for each

Table 10
Project indicators in this study.

Indicator type	Indicator name	Value	Reference
Activity	Number of activities (ACT)	30/60/90/120	
Network	Coefficient of Network Complexity (CNC)	[1.40–2.07]/[1.45–2.08]/[1.47–2.09]/[1.48–2.09]	Vanhoucke et al. (2016); Vanhoucke and Coelho (2018)
	Order Strength (OS)	[0.25–0.64]/[0.20–0.55]/[0.16–0.48]/[0.15–0.43]	
	Serial/Parallel (SP)	[0.17–0.41]/[0.12–0.27]/[0.10–0.20]/[0.09–0.18]	
	Activity Distribution (AD)	[0.11–0.52]/[0.16–0.51]/[0.16–0.50]/[0.19–0.53]	
	Length of Arcs (LA)	[0–0.25]/[0–0.08]/[0–0.04]/[0.01–0.03]	
	Topological Float (TF)	[0.02–0.50]/[0.07–0.50]/[0.10–0.45]/[0.09–0.44]	
Regular resource	Length of long arcs (IS)	[0.55–0.88]/[0.62–0.86]/[0.68–0.84]/[0.69–0.85]	
	Resource Factor (RF)	0.25;0.50;0.75;1	
	Resource Use (RU)	[1–4]/[1–4]/[1–4]/[1–4]	
	Resource Strength (RS)	[0.14–1]/[0.14–1]/[0.18–1]/[0.09–0.52]	
Spatial resource	Resource Constrainedness(RC)	[0.10–0.61]/[0.07–0.49]/[0.05–0.45]/[0.06–0.50]	
	Spatial Resource Constrainedness (SRC)	[0.15–0.32]/[0.13–0.28]/[0.12–0.26]/[0.12–0.27]	

feature, we need to calculate information entropy, information gain and information gain ratio, respectively, detailed calculation formulas can be referred to the study of Hssina, Merbouha, Ezzikouri, and Erritali (2014), Sakthivel, Sugumaran, and Babudevasenapati (2010) and Guo et al. (2021). Thirdly, we will select the feature with the highest information gain rate (the best feature) as root node or internal root, and then branch it based on its feature values. As mentioned earlier, the features we consider are all continuous values and need to be discretized. Finally, once the best feature is selected and the data is partitioned based on that feature, we recursively perform the same process on each partitioned subset until no more features are available for splitting or no more samples are available for partitioning in a certain node, and the branching process is terminated. In addition, we use the error-based pruning (EBP) method to avoid overfitting in the decision tree generated by C4.5 (Sakthivel et al., 2010). Fig. A.3 in the appendix presents the constructed decision tree.

5.8.2. The performance of decision tree method

To verify the effectiveness of the decision tree method, we randomly select 1000 instances from R-instances with a total of 2040, as the training set, while the remaining 1040 instances are used as the testing set. Moreover, the test set of instances on the heuristic algorithm is consistent with the decision tree method. The experimental evaluation metric is designed as follows: we can obtain the best rule from 36 PRs for each instance, which corresponds to the best project makespan. The best rule for each instance could be one or more, and the set of best rules is named as *Best_PR*. Further, for the decision tree method, we automatically detect and predict the best rule for each test instance. If the predicted best rule belongs to the *Best_PR*, it indicates that the decision tree method can be accurately predicted for this instance. Therefore, the percentage of accurately predicted instances out of the total number of test instances is considered as our evaluation metric, denoted as *Percent_Instance*. For each PR, *Percent_Instance* indicates the percentage of projects that return the best makespan to the total testing instances.

Fig. 15 demonstrates the difference in project performance when using the C4.5 method to adaptively select PRs based on project characteristics compared to randomly selecting a PR. Obviously, selecting PRs that are more suitable for a specific project based on their characteristics can lead to better overall performance. We achieved the best rule selection ratio of up to 85% for projects through prediction using our designed decision tree method. However, if we randomly select a rule from the available PRs for experimentation, the proportion of projects with the best makespan is very unstable. Random PR1-PR10 refers to the PRs that are randomly selected. The performance of Random PR1-PR10 ranges from 30% to 77%, which is far inferior to the constructed decision tree model using the C4.5 method. The main reason for this gap may be that although a certain rule performs better for a specific project, it may not necessarily produce better performance

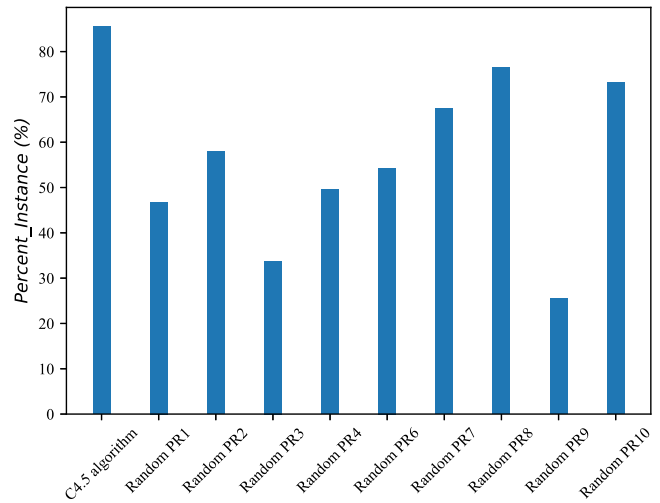


Fig. 15. Comparison of performance between C4.5 and random selection PR.

for all projects. Even the performance achieved by the better rule IC (79%) is inferior to that of the decision tree method. Therefore, we can prove that the constructed decision tree model is effective and can adaptively select rules based on the characteristics of a project.

For each instance, the decision tree method can recommend suitable PRs based on its project's characteristics, and more than one PR may perform well for a certain project instance. In Fig. 16, the vertical axis on the left side demonstrates the number of instances where the decision tree recommends different PRs, and the vertical axis on the right side displays the number of instances obtained from the left vertical axis as a percentage of the total number of testing instances. We can clearly observe that the number of instances is higher when the decision tree method recommends IC and SRCI rules. Although there is not much difference in performance deviation between IC and SRCI in the previous analysis, we find that there is still a significant difference in the number of instances recommended by the decision tree method for a specific rule in Fig. 16. In addition, the decision tree method recommends a particular rule that may be in a relatively better position, but the overall performance of this rule does not match this, and the random rule is a representation of this scenario. The main reason for this phenomenon is that some rules do perform very poorly in some project instances, resulting in an average performance that is not well, but the number of project instances in which it achieves a better result may be a little higher.

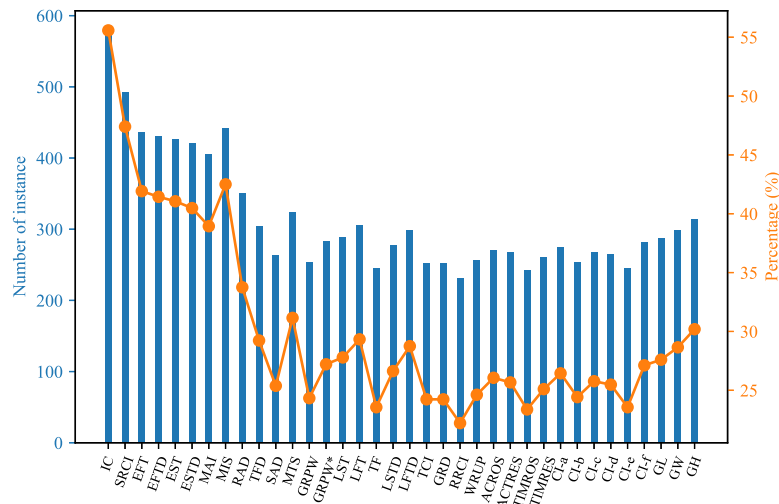


Fig. 16. Number of instances for different PRs recommended by the decision tree method.

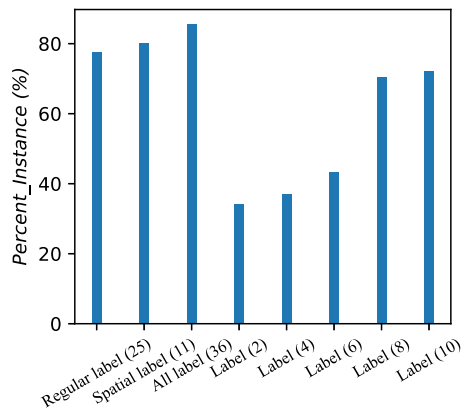


Fig. 17. Performance of decision tree under different labels.

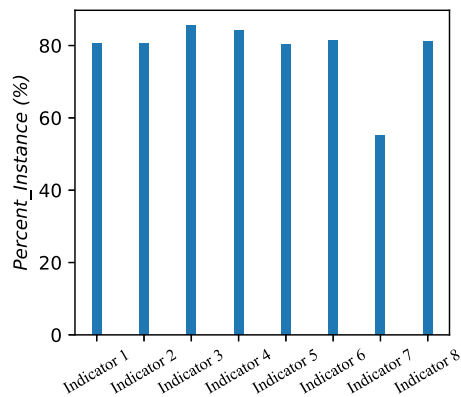


Fig. 18. Performance of decision tree under different indicators.

In addition, Fig. 17 shows the performance of the decision tree method using regular resource rule labels, spatial resource rule labels, all labels, and randomly selecting 2, 4, 6, 8 and 10 labels, respectively. It can be observed that in a mixed state, where both spatial resource rule labels and regular resource rule labels are considered simultaneously, the performance of the decision tree method reaches its best state. In Fig. 18, indicators 1–8 represent “Network indicator”, “Regular resource indicator”, “All project indicator”, “Regular resource+Network indicator”, “Regular resource+Activity indicator”,

“Spatial resource+Network indicator”, “Spatial resource+Activity indicator” and “Spatial resource+Regular resource indicator”, respectively. When we consider all indicators simultaneously, namely activity, network, regular resource and spatial resource indicators, the performance of the decision tree method reaches its best state.

6. Conclusions

In this paper, we propose a new 3D spatial resource-constrained project scheduling problem (3D-sRCPP) to cope with the scenario of time-space conflicts. Two types of spatial constraints are refined to construct the mathematical model for the 3D-sRCPP, which are expressed by non-overlapping among active cuboids and not exceeding the total space, respectively. A specific heuristic algorithm is designed to solve the 3D-sRCPP, which combines multiple PRs and three spatial allocation strategies. The optimal or satisfactory schedule scheme for the 3D-sRCPP can be obtained by our heuristic algorithm. In addition, as an extension study, the decision tree model is constructed, which is adopted to adaptively select the best PRs for each instance according to the indicators of project instances.

Firstly, according to the results from our heuristics, the 36 PRs for the 3D-sRCPP are compared and evaluated based on numerous project instances. We can find that IC (integrative combination) rule performs best, followed by SRCI rule. For all 36 PRs, only IC firstly balances effects among regular resources, 3D spatial resources and time criticality. Moreover, those PRs that perform better for the traditional RCPSP do not act best for the 3D-sRCPP. Secondly, overall, our customized heuristic algorithm is more effective than Gourbi for general instances with medium/large sizes, which can effectively eliminate time-space conflicts in the planning stage of a project. The analysis of the model's effectiveness shows that the average deviation of the project makespan from the 3D-sRCPP is far less than that of actual project durations gained by the temporary response to time-space conflicts in the execution stage. Then, when implementing projects according to the RBSS (ignoring spatial resources), an interesting phenomenon is that the number of time-space conflicts decreases as the breadth of spatial allocation strategies increases. Therefore, the impact of different space allocation strategies on the number of time-space conflicts is worth investigating. Moreover, we can acquire optimal solutions for very small-scale instances by Gourbi within acceptable running time. Finally, our extension research validates the constructed decision tree model, which is the improvement of the heuristic algorithm based on PRs and space allocation strategies. Our research findings can provide

Table A.1
Comparisons of the optimal solution and heuristic solution for each instance.

Category	INS	G	H	GT	HT	DM ₄	INS	G	H	GT	HT	DM ₄	Category	INS	G	H	GT	HT	DM ₄	INS	G	H	GT	HT	DM ₄
OS=0.3 RS=0.25	J15-1	38	43	2852	40	13	J10-1	46	46	114	32	0	OS=0.7 RS=0.25	J15-1	55	55	284	20	0	J10-1	27	27	3	3	0
	J15-2	36	39	1020	22	8	J10-2	41	49	142	13	20		J15-2	63	63	423	21	0	J10-2	44	44	10	4	0
	J15-3	39	39	1102	21	0	J10-3	38	41	147	15	8		J15-3	59	62	504	23	5	J10-3	60	60	42	7	0
	J15-4	39	45	818	14	15	J10-4	35	35	31	12	0		J15-4	77	77	1047	14	0	J10-4	37	37	8	2	0
	J15-5	27	31	391	22	15	J10-5	37	42	101	12	14		J15-5	57	57	144	23	0	J10-5	33	33	4	2	0
Avg	36	39	1236	24	10	39	43	107	20	8	62	63	480	20	1	40	40	13	4	0					
OS=0.3 RS=0.5	J15-1	43	43	33	7	0	J10-1	32	35	60	9	9	OS=0.7 RS=0.5	J15-1	47	47	76	12	0	J10-1	31	34	18	4	10
	J15-2	29	31	216	14	7	J10-2	23	23	27	6	0		J15-2	52	52	32	12	0	J10-2	43	43	14	2	0
	J15-3	23	28	128	26	22	J10-3	31	37	8	2	19		J15-3	45	55	9	2	22	J10-3	32	33	4	5	3
	J15-4	25	25	16	3	0	J10-4	22	22	8	2	0		J15-4	41	41	142	1	0	J10-4	38	38	9	4	0
	J15-5	29	34	652	19	17	J10-5	22	22	10	7	0		J15-5	51	51	113	13	0	J10-5	41	42	7	4	2
Avg	30	32	209	14	9	26	28	23	5	6	47	49	75	8	4	37	38	10	4	3					
OS=0.3 RS=0.75	J15-1	23	23	109	15	0	J10-1	29	30	22	9	3	OS=0.7 RS=0.75	J15-1	37	38	141	16	3	J10-1	40	40	5	2	0
	J15-2	18	18	34	6	0	J10-2	22	22	5	5	0		J15-2	43	45	37	12	5	J10-2	27	27	3	2	0
	J15-3	26	31	28	11	19	J10-3	20	23	26	8	15		J15-3	39	41	10	4	5	J10-3	31	33	20	8	6
	J15-4	30	32	63	21	7	J10-4	16	17	2	4	6		J15-4	30	31	8	3	3	J10-4	46	46	5	3	0
	J15-5	20	20	30	20	0	J10-5	24	24	3	4	0		J15-5	43	49	6	1	14	J10-5	46	46	14	2	0
Avg	23	25	53	15	5	22	23	12	6	5	38	41	40	7	6	38	38	9	3	1					
OS=0.3 RS=1	J15-1	21	21	28	12	0	J10-1	17	17	2	3	0	OS=0.7 RS=1	J15-1	36	36	5	4	0	J10-1	35	35	2	4	0
	J15-2	22	22	21	13	0	J10-2	16	16	3	2	0		J15-2	42	42	7	1	0	J10-2	26	26	1	2	0
	J15-3	27	27	22	16	0	J10-3	22	22	2	1	0		J15-3	35	35	9	2	0	J10-3	35	35	1	1	0
	J15-4	13	13	17	4	0	J10-4	24	24	2	1	0		J15-4	27	27	9	3	0	J10-4	24	24	2	3	0
	J15-5	29	30	32	14	3	J10-5	20	20	3	1	0		J15-5	40	40	11	1	0	J10-5	36	36	1	2	0
Avg	22	22	24	12	0	20	20	2	2	0	36	36	8	2	0	31	31	1	2	0					

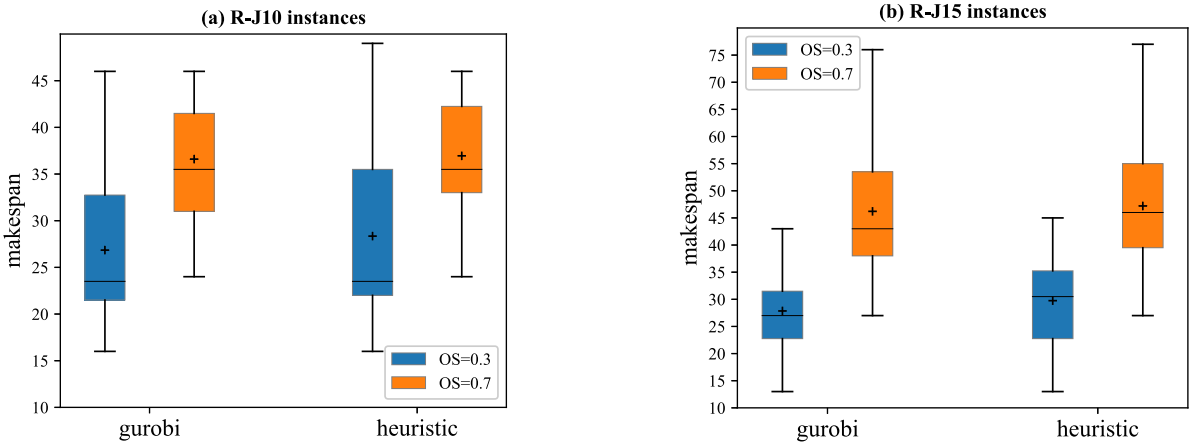


Fig. A.1. Comparisons of optimal solutions and heuristic solutions under different OS.

quantitative guidance for project managers on how to eliminate time-space conflicts in advance when executing a class of large and complex engineering projects in limited construction sites.

The future work of our study mainly focuses on these aspects. On the one hand, we will incorporate our best PRs in meta-heuristics to improve its performance further. On the other hand, projects usually encounter uncertain environments, so we will consider the study of the 3D-sRCPS from the perspective of robust optimization. In addition, we will also explore evolved PRs automatically and compare them with this paper's designed PRs.

CRedit authorship contribution statement

Jingwen Zhang: Supervision, Funding acquisition. **Lubo Li:** Writing – original draft, Methodology, Investigation. **Erik Demeulemeester:** Writing – review & editing, Formal analysis. **Haohua Zhang:** Software, Methodology, Data curation.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

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Appendix A

See [Table A.1](#), [Figs. A.1–A.3](#).

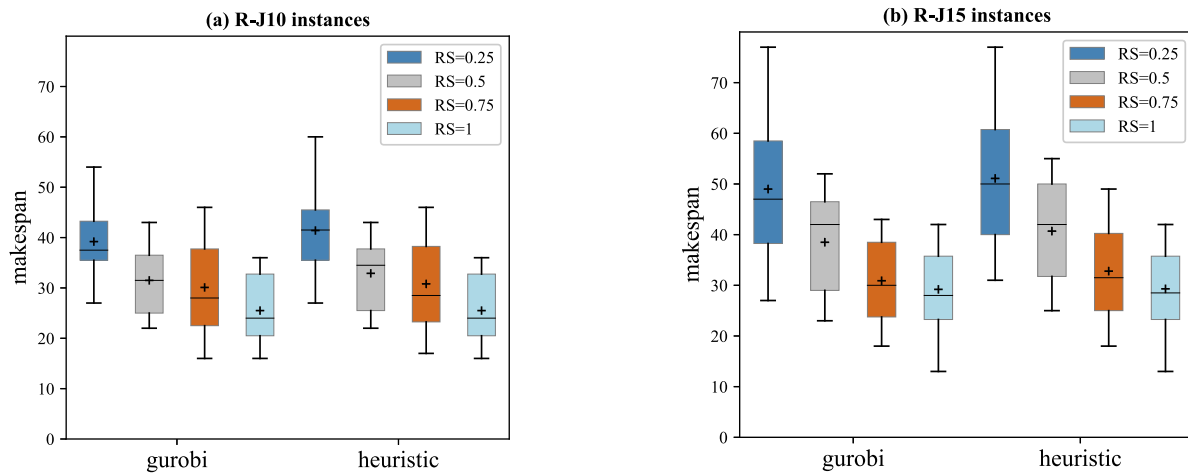


Fig. A.2. Comparisons of optimal solutions and heuristic solutions under different RS.

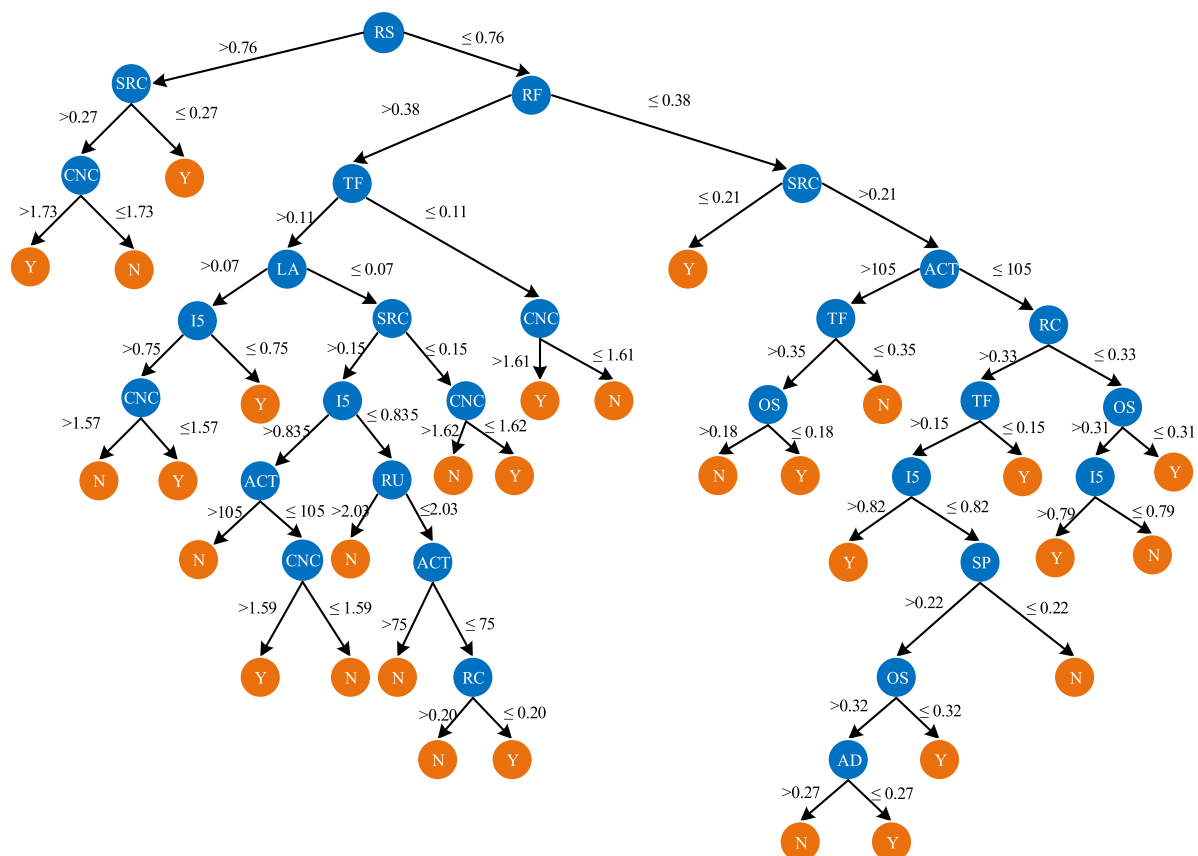


Fig. A.3. The constructed decision tree under one label.

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